LEF Mechanism in Principle

Corotation and angular momentum

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The computational environment table

The calculations of this paper describe an early terrestrial parent body (the proto-Earth), that is, the young Earth before the Moon's mass separated from it, in the LEF scenario. Because the system changes during the process (mass passes from one body to the other, and the shape and the spin change), every quantity below belongs to one of three defined states: the state before the division of mass, a computational reference state used for comparison, or the coupled final state at the moment of release. Keeping these states apart prevents a value from one moment being used at another. Appendix G contains the full environment, E01–E70; the table here lists only the quantities needed for Paper 2.

The table has three columns: the quantity, its numerical value or definition, and its status — an assumption of the system, an input chosen for the model, a value computed in this paper, or a value used only as a sensitivity test. The third column lets the reader see at once which numbers are put into the calculation and which come out of it.

Quantity	Numerical value or definition	Status
Parent mass at the start of the account	6.0455×10^{24} kg — the mass of the Earth and the Moon together	System assumption, full conservation of mass
Parent mass after the transfer	5.972×10^{24} kg	Model input
Transferred mass	7.35×10^{22} kg	Model input
Initial rotation period	5.5 hours	Adopted reference value
Initial volumetric radius	6,517 km	Model input; includes an adopted thermal expansion of 120 km
Initial equatorial radius	about 6,721 km	Computed in the Maclaurin spheroid model

Quantity	Numerical value or definition	Status
Initial mean density	about 5,214 kg/m³	Derived from mass and volume
Initial structure for the budget	approximately homogeneous	Reference assumption, not a conclusion from incomplete differentiation
Final structure in the sensitivity test	two layers; $q = 0.45$–0.65, $D = 1$–3	Computed scenarios
Corotation in the original reference	15,818 km at 5.5 hours, with GM of the final Earth mass in the test-particle approximation	Computational definition; not an insertion of the full initial mass
Earth day and corotation line at release	5.77 hours; 16,331 km (16,399 km two-body), core centralising, $D = 2$	§A.4, §B.1
Release off the equator	25° relative to the spin axis at the moment of release	Sensitivity-test assumption, not a measured ancient latitude; the adopted release latitude is 22.5° (Paper 4)
Body density in the Roche test	2,600 kg/m³	Model input; 1,500 and 3,340 tested as sensitivity
Temperature, volatiles and viscosity	vary between regions; no approved numerical profile	Not replaced by modern values
Gravity	depends on mass, distance and shape in every state	No single gravity figure for the whole process
Duration of construction	not computed in Paper 2	The 600 Myr window is kept in the working scenario of the series, not as a measured time

A few rows need clarification. **The initial parent mass**, 6.0455×10^{24} kg, is the sum of the masses of today's Earth and Moon: no mass is assumed to escape from the system, so the whole Moon was once part of the Earth. **The initial radius**, 6,517 km, is larger than the Earth's radius today, because a young, hot Earth is thermally expanded; the adopted expansion is 120 km. **Maclaurin spheroid:** a spinning fluid body bulges at its equator, and the *Maclaurin model* — the classical equilibrium shape of a spinning fluid body of uniform density, a flattened sphere — turns this volume into an equatorial radius of about 6,721 km through the flattening that the spin produces. **Test-particle approximation:** in the reference state the released body is treated as a small particle in

the gravitational field of the central mass, so that only the Earth's mass enters the formula; the coupled two-body value (16,399 km instead of 16,331 km) shows how small this simplification is. **Structural sensitivity:** in the two-layer rows, q is the ratio of the radius of the dense inner layer to the radius of the whole body, and D the ratio of the inner density to the outer density; $D = 1$ is a uniform body, and D larger means more iron gathered towards the centre. The 25° row is a sensitivity test, used to see how much the result changes when the release does not take place on the equator; it is not the adopted value, which is 22.5° (Paper 4). Where the table says that a quantity has no single value — temperature, viscosity, gravity — the paper does not borrow a modern value in its place. The 600 Myr construction window is the working scenario of the series: about 400 Myr for the bulge and the pear, about 100 Myr of flattening and about 100 Myr to release at 3.873 Ga. Paper 2 does not compute this duration and does not depend on it.

MASS CLARIFICATION

Before the lunar reservoir separates, the mass of the lunar material is wholly included in the mass of the parent body. With the transfer, the same mass is subtracted from the parent and assigned wholly to the second body. The distinction between the total initial mass and the mass used in the reference calculation prevents double counting and the mixing of states.

In practice, the mass of the Moon, 7.35×10^{22} kg, is counted only once. Before the transfer it is included in the parent mass, 6.0455×10^{24} kg; after it, it is the Moon, and the Earth remains with a final mass of 5.972×10^{24} kg, with no loss of mass from the system.

Abstract

The idea. Material that is part of a spinning planet, or still attached to it, turns with it: it completes one turn in every planetary day. This is *corotation*. The tangential speed of such material is the spin rate times the distance from the centre, so it grows the farther out the material is. The speed needed for a free circular orbit behaves the opposite way: it falls with distance. The point where the two speeds are equal is the *corotation radius*. Material carried out to that distance while still turning with the parent, and released there, already moves at orbital speed; no mechanism is needed to throw it.

The question. The paper examines whether transferring lunar material outward from a terrestrial parent, while corotation is maintained, can lead to a state whose orbital velocity and angular-momentum budget approximately match the present Earth–Moon system. (*Angular momentum* is the amount of rotation a system holds; internal processes can only pass it from one part to another, and only outside forces, such as the Sun's tides, can remove it from the system.) The central distinction is between the velocity required at the moment of release and the mechanical capacity and angular-momentum budget needed to reach that state. In a central-field model the corotation radius follows from equating the rotational speed and the orbital speed. For an initial rotation period of 5.5 hours the reference values are a corotation radius of 15,818 km and an orbital speed of 5,020 m/s.

The budget. The parent is described as a uniform Maclaurin spheroid (the flattened equilibrium shape of a spinning fluid body) with a volume-equivalent radius of 6,517 km. The computed initial angular momentum is about $3.466 \times 10^{34}\ \mathrm{kg\,m^2\,s^{-1}}$, a value about 0.7% above the sum of the magnitudes of the present angular-momentum components — a small surplus, which is what the later loss to solar tides requires. Vector addition at the angles tested gives a surplus of about 1.5–2.5%; this is a sensitivity test, not a full vector reconstruction of the evolution. The budget depends on the nearly uniform structure and on the adopted thermal expansion.

The release. Two opposing processes change the Earth's spin during the transfer of mass: the lengthening pear carries angular momentum outward and slows the spin, and the centralising core — iron sinking inward — speeds it up. The two act together; with $D = 2$ the Earth's rotation period at release is 5.77 hours, and the corotation line moves out to about 16,400 km, where the Moon's centre of mass is released. The angular momentum transferred to the lunar material is about 15.6% in the base calculation and about 16.5% in the coupled solution. The final orbital angular momentum is about 17.7% of the initial budget in the coupled solution with the pear alone ($D = 1$), and about 17.0% in the release state with $D = 2$ (§B.1). These are transfers within the system.

The result. The paper establishes the kinematic solution and the budgetary end state of the route: the speed a corotating mass needs is supplied by the rotation itself at the corotation radius, and the angular momentum the transfer requires is available in the parent. What the route then requires of the mechanism — that the body be released beyond the corotation radius, that it survive the years it spends inside the fluid Roche limit, and that the connection carry the load until then — is stated as a set of explicit conditions in Part D. Paper 3 examines those conditions.

The layout. Part A defines the constraints the route must meet; Part B solves them in the same order; Part C summarises the quantitative findings; Part D states claims, tests and requirements; Part E collects the conclusions. Appendices A–G hold the full mathematical derivations and the full computational environment.

WHAT THIS PAPER PROPOSES

The Moon's material left the Earth while still turning with it, and was released where the speed it already had was the speed a free orbit requires. That place is the corotation radius: 15,818 km for a parent with a rotation period of 5.5 hours, at a speed of 5,020 m/s. The objection that led to the abandonment of the fission route — an angular-momentum budget four times larger than required — was evaluated at the parent's surface, under a release condition this route never uses. Evaluated at the corotation radius, the parent's budget exceeds that of the present system by about 1%, with the sign that solar tidal loss requires, and the classical route needs 2.74 times what this route needs.

> **The route states its threshold conditions explicitly:** a parent still at an early stage of sorting at the start of the transfer of mass ($k \gtrsim 0.397$), thermally expanded by 98–520 km; a body released beyond the corotation radius, and not exactly on it; an interior stiff enough to survive 10^1–10^3 years inside the fluid Roche limit; and a connection able to carry the load until then. Each of these is a defined requirement, each is passed to the paper that supplies it, and all of them are listed in §D.3. The requirements of the model, and what would test it, are collected in Paper 11 (The Requirements).

Why the old objection does not apply. The classical fission model assumed that the Earth throws material directly off its own surface. At the surface of a parent with a rotation period of 5.5 hours, the tangential rotational speed is only about 2,133 m/s, while the speed needed for a free orbit at the same place is about 7,701 m/s (§A.1.2). Closing the velocity gap at the surface requires a much faster spin of the whole Earth, and a faster spin means more angular momentum: at the classical point where a spinning fluid body becomes unstable (the Jacobi bifurcation, §A.6.5) the parent needs 2.74 times the angular momentum of the present Earth–Moon system, and the earlier reference calculation, made at another point of the same sequence, indicated a requirement of about four times. The system does not have such angular momentum, and this is the finding that led to the abandonment of the fission theory. The LEF route closes the gap in a different way: it does not increase the Earth's spin, but carries the material outward while it turns with the parent. The tangential speed grows with distance at the same spin rate, until it equals the orbital speed at the corotation radius. The spin rate needed for this is the one the Earth would have had if all of the system's present angular momentum were returned to it — about 5.5 hours (§A.2) — and the angular-momentum budget then needed is only 1.007 times today's value. An everyday picture: a child on a roundabout who walks from the centre to the rim moves faster at the rim, without anyone pushing the roundabout harder.

What the requirements mean. Mass distribution: k is the moment-of-inertia coefficient, and it measures how the mass of a body is spread between centre and surface. A uniform homogeneous sphere has $k = 0.4$; today's Earth, with its iron core, has about 0.33. The model requires $k \gtrsim 0.397$ at the start of the transfer of mass, that is, a parent still at an early stage of sorting: mass that is not concentrated toward the centre stores more angular momentum at the same spin rate. The organisation of the Earth's iron core is one continuous process that never stopped: during the bulge stage it was at its beginning, it continued while the pear grew, and it went on after release, while the Earth took the pear back in. There was probably never a fully uniform period: the earlier the stage, the less sorted the Earth. Iron may already have begun to gather in the core at the start; the process continued. **Thermal expansion:** the young Earth must be hot, and therefore larger than today: an expansion in the range 98 to 520 km is admissible, and the adopted working value is 120 km. **Release margin:** the release must take place *beyond* the corotation radius and not exactly on it. On the line itself the tide is neutral and pushes in neither direction, while a body lying even slightly beyond the line is pushed outward by the tide, so that it recedes from the start (Paper 4); the margin needed is small, $\Delta r \simeq 2d_0$, about 210 km (Paper 3, §C.5a). **Survival at the Roche limit:** the *fluid Roche limit* is the distance inside which the Earth's tidal forces would pull apart a body held together by its own gravity alone. A body with internal strength survives closer in, down to its rigid limit; the

Moon at release is a rigid, round body, having completed its flattening and rounding earlier, while still attached to the parent (Paper 7). **Load transmission:** the connection between the two bodies — the pear — must carry the load until the moment of release. These requirements are not hidden in the results; they are defined explicitly in §D.3, and Paper 3 tests them.

Introduction — Part A: the release condition in the context of existing models

Two kinds of evidence bear on the Moon's origin: how its material could have reached orbit (the dynamics of separation), and whether its composition matches the body it came from. Darwin's work on tides and rotation is part of the historical background for examining the evolution of the Earth–Moon system [6]; his fission idea — that a fast-spinning Earth shed the Moon — is the classical form of a terrestrial origin. In this paper the quantitative comparison with classical fission is made through a defined equilibrium configuration. The bounds derived from this configuration hold for it alone, and must not be turned into a bound on every possible path of terrestrial separation. In other words, showing that one particular fission path needs too much angular momentum does not show that every path from the Earth does.

In the giant-impact family of models, the Moon forms from debris after another body strikes the Earth, and the isotopic composition places constraints on the mixing of terrestrial and impactor material. Different versions of the model exist, and the constraints depend on the compositions and on the manner of mixing [7]. In the LEF scenario, construction from a shared terrestrial reservoir is examined: the Moon is made of the Earth's own material, so a close isotopic match follows directly. A shared source provides a basis for compositional comparison, but does not replace the test of transfer to orbit. That test is the subject of this paper.

The focused question of this paper is whether carrying mass outward in shared rotation changes the velocity condition and the angular-momentum budget relative to direct release from the surface. The answer the paper gives is positive: the release condition moves from the surface to the corotation radius, and both the speed and the budget then fit. The mechanism of construction — the feed from within and the rise of the outer shell as the surface of the same body — is the subject of Paper 3, *LEF Mechanism in Practice*. The present paper defines the physical conditions the mechanism must meet. The division is deliberate: Paper 2 examines the feasibility of the route in principle, using standard physics; Paper 3 shows how it is carried out.

Introduction — Part B: what the paper presents

The paper is built as a set of problems followed by their solutions, so that each objection is met in the place where it is raised. Part A sets out the constraints the route must meet, one by one: the tangential velocity gap at the surface, which led to the rejection of classical fission; what must be calculated; the pear that connects the Earth to the corotation line; the offset between the lengthening pear, which slows the spin, and the centralising core, which speeds it up; corotation

itself; the angular-momentum budget; the fluid Roche limit; and the direction of tidal evolution. Part B solves them in the same order, starting from one calculation. Part C summarises what has been shown; Part D collects the claims, tests and requirements; Part E states the conclusions. Appendices A–G give the mathematical derivations, the numerical sensitivity scans, the Roche calculations and the parameters of the full computational environment, so that every number in the main text can be traced and reproduced.

> **THE EQUATORIAL PLANE**
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> For computational simplicity and clarity of principle, the calculation and the whole of this paper treat the release of the Moon from the equatorial plane. Paper 4 shows that the Moon left from another latitude — 22.5, the centre of the contact area — and computes the release there in full.

The kinematic analysis of this paper is carried out in the equatorial plane, defined as the simplest geometric case: there the tangential rotational speed is greatest, and the corotation radius is a single distance. Moving the release to a higher latitude changes the numerical values to a limited extent, without changing the mechanical principle. The size of the change is shown by the 25° release sensitivity test of the environment table, and the adopted release solution, at a latitude of 22.5°, is computed in full in Paper 4.

PART A

The problems before us

> **THE PROBLEM AND HOW IT IS TESTED**
>
> Each section of this Part states one mechanical condition that a body built from the Earth's own material must meet if it is to reach an independent orbit whole. Part B answers these conditions in the same order.

This Part breaks the dynamical problem into a series of successive physical conditions that the route must meet. It begins with the difficulty that closed the classical fission route — a body leaving the surface of a fast-spinning Earth does not have the speed of an orbit — and then examines, one after another: where the matching orbital speed is reached; what mechanical connection joins the Earth to that place; how the parent's spin changes while the mass is carried out; whether the total angular-momentum budget suffices; whether the body withstands the Earth's differential tide (the Roche limit); and what the direction of the orbit's tidal evolution is afterwards. Each section yields a defined quantity, which Part B computes.

A.1 The problem: the velocity gap

A.1.1 Two separate dynamical questions

A terrestrial-origin scenario requires an explanation for the transfer of material from the parent body to a free orbit. A compositional match can support a shared source, but does not determine how the material reached orbit: two bodies may be made of the same material and still leave open the question of how one of them came to circle the other. The focus of this paper is therefore the dynamics. The original historical review is examined here by comparing defined equilibrium configurations, and not by asserting that every terrestrial-origin path is subject to the same fission threshold.

Analysing such a scenario requires two mechanical questions to be kept apart. The first is the release-velocity condition: under what conditions the velocity of the material at the moment of separation is consistent with a free orbit — that is, whether, at the moment it is let go, the material moves fast enough sideways to keep falling around the Earth without falling back onto it. The second is the angular-momentum budget: whether the angular momentum available at the start of the process permits the dynamical end state and the evolution that follows. Changing the release condition can change the velocity test, but the overall angular-momentum budget must still be satisfied. A route can pass the first test and fail the second, so both are computed.

Angular momentum and the moment of inertia. Angular momentum is the quantity of rotational motion a body carries. For a spinning body it is the product of the spin rate and the way the mass is spread around the axis. The spin angular momentum is defined by the equation:

$$L_{\mathrm{spin}} = I\omega.$$

where ω is the angular velocity (the angle the body turns per second, 2π divided by the period), and I is the moment of inertia: the sum, over every part of the body, of the mass of each part times the square of its distance from the spin axis. The moment of inertia measures how far from the axis the mass sits; mass far from the axis counts much more than mass near it, because the distance enters squared. For a body of mass M and equatorial radius a the moment of inertia is written in the form $I = kMa^2$, where the dimensionless coefficient k defines the internal mass distribution: for a uniform body $k = 0.4$, and for a body with a dense core k is smaller.

In the absence of an external torque — a twisting force acting on the system from outside — the total angular momentum is conserved: it cannot change, but only pass from one part of the system to another. In the Earth–Moon system the only significant external torque is the tide raised by the Sun, and it has removed only about 1% of the total angular momentum over the history of the system (§A.2). This conservation is what makes the problem calculable: whatever the Moon carries in its orbit today was once in the spin of the parent.

Transferring mass outward in corotation — that is, while it keeps turning with the parent at the parent's own angular velocity — requires the parent to supply additional angular momentum to the transferred material, because the same angular velocity at a larger distance means a larger moment of inertia for that material, in proportion to the square of the distance. Since there is no external

torque in this process, it is a redistribution of existing angular momentum: the angular momentum gained by the transferred material is subtracted directly from the parent's spin.

A.1.2 Release from the surface

We first examine the classical case: mass let go directly from the equator of a fast-spinning parent. In the reference state the period is $T_0 = 5.5$ hours and the equatorial radius is $a_0 \simeq 6,721$ km. Since 5.5 hours are 19,800 seconds, the angular velocity of the parent is:

$$\omega_0 = \frac{2\pi}{19,800} \simeq 3.17333 \times 10^{-4}\,\mathrm{rad\,s^{-1}}.$$

Two speeds are now compared at the equator. The first is the tangential speed that the rotation itself gives to a point on the surface, v_{rot}: the angular velocity times the distance from the axis. The second is the circular orbital speed v_{circ}: the speed at which a free body at the same distance would keep falling around the Earth in a circle, its centrifugal tendency exactly balancing gravity.

$$v_{\mathrm{rot}} = \omega_0 a_0 \simeq 2,133\,\mathrm{m\,s^{-1}}, \qquad v_{\mathrm{circ}} = \sqrt{\frac{GM_\oplus}{a_0}} \simeq 7,701\,\mathrm{m\,s^{-1}}.$$

Even with a day of only 5.5 hours, the parent's surface speed is less than a third of the speed an orbit at that place requires. The velocity difference, in the rounding of the original reference calculation of this paper, is about 5,568 m/s. This is the velocity gap.

What orbit would such material follow if it were simply let go? Without an additional radial velocity, the specific orbital energy (the orbital energy per kilogram, kinetic minus gravitational) and the semi-major axis (half the long axis of the resulting ellipse) are:

$$\varepsilon = \frac{v^2}{2} - \frac{GM_\oplus}{r} \simeq -5.70 \times 10^{7}\,\mathrm{J\,kg^{-1}}, \qquad A_{\mathrm{orb}} = -\frac{GM_\oplus}{2\varepsilon} \simeq 3,495\,\mathrm{km}.$$

The negative energy indicates strong gravitational binding, and the semi-major axis, 3,495 km, is about half the parent's equatorial radius. Since the release point is the farthest point of the ellipse (the apoapsis), the whole of the rest of the path lies deep inside the parent. Under these initial conditions the Keplerian orbit penetrates the parent, and the material falls back. Surface velocity alone therefore does not provide a free orbit that remains outside it. This is the core of the classical objection to fission: closing the gap at the surface requires the parent to spin several times faster, and this requires far more angular momentum than the Earth–Moon system holds (§A.6.5).

The numbers in §A.2 preserve the original reference comparison, in which GM of the final Earth mass was used in the test-particle approximation. They are not a calculation of the exact field of the whole initial parent. This definition distinguishes the reproduced reference comparison from the overall budget in §A.6.

A.1.3 The transfer condition examined

The LEF route does not release the material at the surface. In the scenario examined, the material is carried outward while still attached and receives angular momentum, so that it keeps turning with the parent at a uniform angular velocity. The two speeds then behave in opposite ways. The tangential speed ωr grows with distance, because a point that completes one turn per day travels a larger circle the farther out it is — like a child who walks outward on a turning roundabout and is carried round faster and faster. The required orbital speed $\sqrt{GM/r}$ decreases, because the gravitational field weakens with distance and less sideways speed is needed to balance it. The velocity gap of §A.1.2 therefore narrows continuously as the material rises, and at a certain distance — the point of equality — it closes.

The present paper concentrates on computing the point of equality and the angular-momentum cost of the transfer; the ascent process itself is a separate mechanical question. Paper 3, *LEF Mechanism in Practice*, examines the transport capacity and the connection; the detailed pressure calculation lies outside the budget calculation presented here. The orbital initial conditions after separation are handed to Paper 4, *The Orbit*.

A.2 What must be calculated

The problem now has a definite shape. The model defines five quantities required for the solution: the release distance of the Moon; the length of the Earth day at the moment of separation; whether the available angular momentum suffices; whether the body survives the differential tidal forces (the Roche test); and whether it recedes after release.

Why 5.5 hours. The initial day is not a free variable; it follows from conservation of angular momentum run backwards in time. All of today's Earth–Moon angular momentum, 3.44×10^{34} kg m^2 s^{-1} — the Earth's spin together with the Moon's orbit — is returned to the spin of a young, hot, expanded Earth still at an early stage of sorting ($k = 0.4$ as the calculation input). "Expanded" means that the hot early Earth was somewhat larger than the cold Earth of today; "an early stage of sorting" means that the organisation of the core had only begun, so the mass was spread nearly uniformly and the uniform value $k = 0.4$ describes it closely. Putting the whole budget into the spin of such a body gives the period:

$$T_0 = \frac{2\pi I_0}{L} \simeq 5.54\,\mathrm{h}.$$

In words: a body with moment of inertia I_0 carrying angular momentum L turns at $\omega = L/I_0$, and its period is $2\pi/\omega$. Solar tides have since removed about 1% of the system's total angular momentum (about 1.35% when the receding Moon is followed; Eulenfeld & Heubeck 2023; Canup 2004). A period of 5.5 hours therefore carries today's angular momentum and the roughly 1% that was lost: it is slightly shorter than 5.54 hours because the early system held slightly more. Assuming a young Sun more massive by a few per cent raises this loss only to 1–2%.

A.2.1 The reference state

For $GM_{\oplus} \simeq 3.986 \times 10^{14}\ \mathrm{m^3\,s^{-2}}$ and $T_0 = 5.5$ hours, the point of equality at which the tangential rotational speed equals the orbital speed defines the corotation radius (derived in §A.5), and the common speed there:

$$R_{\mathrm{sync},\,0} \simeq 15,818\ \mathrm{km}, \qquad v_{\mathrm{sync},\,0} \simeq 5,020\ \mathrm{m\,s^{-1}}.$$

This is the central number of the paper. At 15,818 km, more than twice the parent's equatorial radius, a body that completes one turn every 5.5 hours with the Earth already moves at 5,020 m/s, which is exactly the speed of a circular orbit at that distance. A body held there in corotation and then let go does not need any additional push: at the moment of separation it is already in orbit.

The distance is measured from the centre of the central body in the test-particle approximation, which neglects the orbiting body's own mass in the gravitational pull. Substituting this radius into the two velocity formulas confirms that the speeds are equal.

The derivation is an application of the equations of motion in a central field [8]. If an initial state is computed with the full parent mass, M_{tot} must be inserted in the formula. The value 15,818 km remains the reference point defined in the table, and the coupled solution in §A.4 explicitly includes the division of mass.

A.3 The pear

The corotation radius lies far above the surface, so something must carry the material there while it keeps turning with the Earth. In the model, the transport of mass is carried out by a continuous material bridge called "the pear": a hill that rises from the Earth, with a base about 9,000 km wide at the surface, sides that form a single arc without a narrowing (a waist), and at its tip a contact disc about 3,000 km wide at the Moon.

The pear bridges about 7,800 km between the Earth's surface and the Moon. It lies entirely inside the corotation line, so it is pulled inward: at every point along it, gravity is stronger than the acceleration the rotation needs to keep the material circling, so the pear's own material is held by the Earth and does not fly off. Only the body at its tip reaches the line. The pear's length and mass are a requirement: the Moon was released as a whole unit, so a pear must exist that meets the conditions. How the pear is built and how it carries the load are the subject of Paper 3.

Figure 1. The pear laid on the drawing, to scale: Earth at a 5.77-h spin, the Moon with its centre on the corotation line.

Figure 1 shows the proportions to scale: the pear's broad base on the Earth, the single arc of its sides, and the Moon at its tip with its centre on the corotation line. The day of 5.77 hours used in the drawing is the length of the day at the moment of separation, computed in §A.4.

A.4 The offset: two processes acting together

To determine the length of the Earth day at release, two processes that act on the Earth's spin at the same time, and in opposite directions, must be taken into account. This matters because the day at release fixes the position of the corotation line, and therefore the place where the Moon is released.

1. The pear lengthens. Transporting mass outward with the rotation carries angular momentum with it; the angular momentum that leaves the Earth slows its spin. A child walking outward on a roundabout slows the roundabout in the same way.

2. The core centralises. Differentiation is the separation of an initially mixed body into layers by density: iron, the densest component, sinks toward the centre and forms a core, while the lighter materials remain above it. In the LEF model the organisation of the core is one continuous process: it began before the transfer, continued while the pear grew, and went on after release. Fast rotation speeds up differentiation: the inward migration of iron reduces the Earth's moment of inertia, because mass moves closer to the axis. With a smaller moment of inertia and the same angular momentum, the body must turn faster. As with a skater drawing in her arms, the spin speeds up.

Angular momentum is conserved, and the two processes offset each other: the result is the offset between them. A calculation that takes only one of them into account does not describe the Earth of that time — the pear alone, without differentiation, would give too long a day, and the core alone, without the pear, too short a one.

Process	Effect on the length of the day	Shift of the corotation line
1. The pear lengthens	$+0.94$ h (alone: 5.50 $\rightarrow$ 6.44 h)	outward (alone: 15,818 $\rightarrow$ 17,569 km)
2. The core centralises ($D = 2$)	-0.67 h	inward
Both together	**5.50 $\rightarrow$ 5.77 h**	**15,818 $\rightarrow$ 16,331 km**

The table reads row by row. The pear alone would lengthen the period by 0.94 hours and move the corotation line outward by about 1,750 km, because a slower spin needs a larger radius before the rotation supplies orbital speed. The centralising core offsets 0.67 hours of this slowing. The combined result is a modest lengthening of the day of only 0.27 hours, from 5.50 to 5.77 hours, and the line moves out only modestly, to 16,331 km.

Figure 2. The offset: the lengthening pear adds 0.94 h; the centralising core removes 0.67 h.

Figure 2 shows the same balance graphically: the bar added by the pear, the bar removed by the core, and the net result between them.

Figure 3. The corotation line during the transfer: the pear alone, and together with the centralising core.

Figure 3 follows the corotation line as the transfer proceeds. With the pear alone the line runs steadily outward; with the core centralising at the same time it stays much closer to its starting position.

A.4.1 The effect of differentiation on the final state

Concentration of mass inward changes the moment of inertia. To test this, the two layer boundaries, the density, the shape and the spin were computed together. This is an extension of the sensitivity test, and not merely a replacement of k inside a fixed homogeneous figure: a body that differentiates also changes its flattening, and the flattening in turn changes the moment of inertia, so all four must be solved at once. The boundaries were allowed to depart from a spheroidal shape; the limitation of the spheroidal assumption in layered systems is also discussed in the literature [5].

The layered structure profile is defined by two dimensionless parameters. The volumetric radius ratio q is the ratio of the radius of the dense inner layer to the total radius of the body; $q = 0.55$ places the layer boundary a little beyond half the radius, comparable to the present Earth's core. The density ratio $D = \rho_{\text{inner}}/\rho_{\text{outer}}$ is the ratio of the density of the inner layer to that of the outer layer: $D = 1$ is a uniform body, $D = 2$ an inner layer twice as dense as the outer, and higher values represent more advanced differentiation. The larger D is, the more mass sits close to the centre and the smaller the moment-of-inertia coefficient k (for $q = 0.55$ and $D = 2$, $k = 0.360$; §B.6). For a volumetric radius ratio $q = 0.55$, and a density ratio D, the solution of the system in the test-particle approximation gives:

Density ratio D	Final period, hours	Final corotation radius, km
1	6.438	17,569
1.5	6.077	16,907
2	5.770	16,331
2.5	5.505	15,827
3	5.274	15,382

The values show one consistent trend: as D rises from 1 to 3, the final day shortens from 6.438 to 5.274 hours, and the corotation line moves inward from 17,569 to 15,382 km. The first row, $D = 1$, is the pear acting alone, without differentiation; the row $D = 2$ is the adopted state of §A.4. At $D = 2.5$ the speed-up of the spin caused by iron sinking inward almost entirely cancels the slowing caused by transporting the pear's mass, and the day returns to a value close to its initial one.

Differentiation in the final structure can offset part of the spin-down that accompanies the transfer. The rows of the table represent possible states of the system, and do not describe a chronological sequence of evolution in time. All rows rest on the same initial budget of a homogeneous parent;

they do not establish that a concentrated initial structure would provide that same budget. The assumptions, the full scan and the convergence checks are detailed in Appendix C and in the calculation document linked to it.

A.5 Corotation

This section derives the corotation radius used throughout the paper, and defines the domain in which its use is valid.

Figure 4. The speed carried by the rotation and the orbital speed at a 5.77-h day; they cross at the corotation line.

Figure 4 plots the two speeds of §A.1.3 against distance from the Earth's centre. The speed carried by the rotation rises in a straight line, while the orbital speed falls. Inside the crossing point the rotation supplies less than an orbit requires, and the material is pulled inward; outside it the rotation supplies more than an orbit requires, and the material tends outward. The crossing point is the corotation line.

Point of application: the laws of corotation and of gravitational balance apply to the body's centre of mass, not to its geometric centre, because the balance between gravity and rotation applies to the body as a whole, and the whole body moves as if its mass were gathered at its centre of mass. The Moon's centre of mass is displaced slightly toward the Earth (in today's system the displacement is about 2 km), so the corotation forces may have pulled the Moon slightly after its geometric centre crossed the line: the geometric centre crosses the corotation line first, and only after it the centre of mass. Release takes place slightly after the centre of mass crosses the line, about 210 km beyond it (§D.3; Paper 3, §C.5a).

A.5.1 The derivation

In the test-particle approximation, in the equatorial plane and in a central gravitational field, the speed carried by the rotation is the angular velocity times the distance. The circular orbital speed follows from setting the gravitational pull equal to the force needed to keep a mass m on a circle of radius R:

$$v_{\mathrm{rot}} = \omega R, \qquad \frac{GMm}{R^2} = \frac{m v_{\mathrm{circ}}^2}{R}, \qquad v_{\mathrm{circ}} = \sqrt{\frac{GM}{R}}.$$

Since the mass of the orbiting body m cancels from both sides of the equation, the orbital speed at a given distance is the same for any body — for a grain and for a Moon alike. Equating the speeds gives the relation:

$$\omega R = \sqrt{\frac{GM}{R}} \quad \Rightarrow \quad \omega^2 R^3 = GM.$$

Solving for R, and substituting $\omega = 2\pi/T$, gives:

$$R_{\text{sync}} = \left(\frac{GM}{\omega^2}\right)^{1/3} = \left(\frac{GMT^2}{4\pi^2}\right)^{1/3}$$

The equation shows that the corotation radius is a function only of the mass of the central body and of its own rotation period, and grows as the period to the power two-thirds: a slower spin pushes the line outward. Hence a slowing of the spin (the lengthening of the day by the pear) moves the line outward, while a speeding of the spin (the centralising core) brings it back inward (§A.4).

For a positive, constant mass and angular velocity there is a single positive root. This is the synchronous-orbit relation of standard orbital mechanics — the same relation that places geostationary satellites above today's Earth. What is new in LEF is not the relation itself but the place where it is evaluated: the classical objection evaluated the speed condition at the parent's surface, while the LEF route evaluates it at the corotation radius.

A.5.2 The release condition and the domain of validity

Equatorial orbital criterion: the equality fits an equatorial circular orbit provided the radial velocity is zero and the contact forces cease — that is, the body moves purely sideways, and the connection no longer pushes or pulls on it. When there is additional velocity, or when the release takes place off the equator, the initial-condition vectors must be computed. The orbital detail belongs to Paper 4; in this paper only the correction needed for the angular-momentum budget is kept, in §A.6 and Appendix D.

Energy and momentum: reaching the corotation radius does not cancel the work required to lift the material or to transfer angular momentum to it: the speed condition is met at the line, but energy and angular momentum still have to be delivered to the material along the way.

Corrections to the central field: the analytical description is simplified. The field of a flattened parent, the size of the developing body and two-body gravity add corrections to the central description. The coupled solution (§A.4) includes a partial two-body correction, but is not a full solution of two deformed figures connected to each other.

A.5.3 The meaning of the condition for release

The coupled solution in §A.4 defines a budgetary end state: how much angular momentum is in the Earth's spin and how much in the Moon's orbit at the moment of release. It does not by itself determine the onset of recession. An added radial velocity is also not a sufficient condition for sustained recession, because it changes the eccentricity (how elongated the orbit is) and the tides. The lag model and the rheology (the description of how the material deforms and flows under load) must be mutually consistent [3,4]. The direction of the tidal push on each side of the line is set out in §A.8.

A.5.4 The boundary between Paper 2 and Paper 4

Paper 2 fixes the mass, the spin and the angular-momentum budget handed over as the end state for the orbital test. This paper does not compute a recession time, and does not impose a general prohibition on a free orbit inside the corotation radius. **Paper 4** receives the budget data (§A.6 and Appendix D) and treats the orbital elements, the tilt and the evolution after release.

The connected state: in the connected scenario, the free-body equations are replaced by the equations of the connected state where they cease to apply: while the Moon is attached to the Earth through the pear, it is held by the connection and is not a free body.

A.6 The angular-momentum budget

The velocity gap is solved at the corotation radius. The second question of §A.1.1 remains: does the parent's angular-momentum budget suffice to place the Moon at the release point while remaining consistent with the system we see today? This section builds the budget step by step: the parent's initial angular momentum, the cost of carrying the material out, the comparison with the present system, the sensitivity to period, expansion and structure, and the comparison with the classical fission route.

A.6.1 The parent and the initial budget

The initial parent mass is the sum of the masses of the Earth and the Moon, $M_{\rm tot} = M_\oplus + m$. At this stage the material to be transferred is still included in the parent's moment of inertia. The body is described as a Maclaurin spheroid: the equilibrium shape of a uniform fluid body in steady rotation, flattened at the poles and bulging at the equator, symmetric about its axis; the faster it spins, the flatter it is. In the homogeneous Maclaurin model, under the assumptions of the reference state, the following values are obtained [1]:

$$I_0 = \frac{2}{5} M_{\rm tot} a_0^2 \simeq 1.092 \times 10^{38}\ \mathrm{kg\,m^2},$$

$$L_0 = I_0 \omega_0 \simeq 3.466 \times 10^{34}\ \mathrm{kg\,m^2\,s^{-1}}.$$

The first line is the moment of inertia of a uniform spheroid ($k = 2/5$) with an equatorial radius of 6,721 km; the second multiplies it by the angular velocity of a 5.5-hour day. The resulting value, 3.466×10^{34}, is the whole angular-momentum reservoir the route has to work with. The shape equation and the full substitution are detailed in Appendix B. The assumption of a uniform body serves as the reference; changes of structure are examined below within the sensitivity analysis.

A.6.2 The cost of the transfer in the base calculation

Before the transfer, the lunar material is part of the parent, turns with it and carries its share of the parent's angular momentum. Assuming a uniform mass distribution, the mean squared distance of the material from the spin axis is the same as that of the whole body, and its share of the angular momentum is:

$$L_{0,\,\mathrm{mat}} = m\omega_0 \left\langle r_\perp^2 \right\rangle = \frac{2}{5} m\omega_0 a_0^2 \simeq 4.21 \times 10^{32}\ \mathrm{kg\,m^2\,s^{-1}}.$$

Here $\langle r_\perp^2 \rangle$ is the average, over the material, of the squared distance from the spin axis. When the same mass is carried out to the corotation radius and still turns with the parent, its angular momentum grows with the square of the distance from the axis:

$$L_{\mathrm{orb},\,0} = m\omega_0 R_{\mathrm{sync},\,0}^2 \simeq 5.83 \times 10^{33}\ \mathrm{kg\,m^2\,s^{-1}},$$

This value is about fourteen times higher, because the distance from the axis has grown and enters squared. The difference between the two must be supplied by the parent's spin. As a fraction of the whole budget:

$$\frac{L_{\mathrm{orb},\,0} - L_{0,\,\mathrm{mat}}}{L_{\mathrm{initial}}} \simeq 15.6\%.$$

The transfer therefore requires about one sixth of the parent's angular momentum to be handed to the material. This is the amount by which the pear slows the parent's spin (§A.4), before the centralising core is taken into account.

Sensitivity to the source of the material: in the base calculation, an equatorial surface source requires about 13.8%; a representative sample of a similar interior region, at half the original radius, requires about 16.5%. The source matters because material that starts far from the axis already carries more angular momentum, and so needs a smaller addition. A specific interior source requires its own mean squared distance; the word "interior" must not be taken to imply that the global mean necessarily applies to it.

A.6.3 Comparison with the present system

The parent's budget must now be compared with the angular momentum the Earth–Moon system holds today. The present total has two components: the Earth's own spin and the Moon's orbit. The reference constants, copied from the source, give:

$$L_{\mathrm{spin}} = I_\oplus \omega_\oplus \simeq 5.86 \times 10^{33},$$

$$L_{\mathrm{orbit}} = \mu_{\mathrm{red}} \sqrt{G M_{\mathrm{tot}} a_m \left(1 - e_m^2 \right)} \simeq 2.855 \times 10^{34}\ \mathrm{kg\,m^2\,s^{-1}},$$

where $a_m = 3.844 \times 10^8$ m and $e_m = 0.0549$. The second formula is the orbital angular momentum of two bodies circling their common centre of mass: $\mu_{\mathrm{red}} = M_\oplus m / \left(M_\oplus + m \right)$ is the reduced mass, a_m the semi-major axis of the lunar orbit and e_m its eccentricity. The scalar sum of the components is about 3.441×10^{34}, of which about 83% is stored in the orbit: most of the system's angular momentum now lies in the Moon's orbit, not in the Earth's spin. The surplus of the parent's budget relative to this sum is about 0.7%. The parent therefore held slightly more than the system holds today, and this is the sign required by the small loss to solar tides (§A.2).

Vector addition: the spin angular momentum and the orbital angular momentum are vectors that do not point in the same direction, because the Earth's axis is tilted against the plane of the lunar orbit. If the angle between the vectors is θ, the total magnitude is:

$$L_{\mathrm{vec}} = \sqrt{L_s^2 + L_o^2 + 2L_s L_o \cos\theta}.$$

Adding two arrows at an angle gives a shorter total than laying them end to end in a straight line; so the vector total is smaller than the scalar sum, and the parent's surplus over it is larger. For the test angles 18.3°, 23.4° and 28.6° one obtains surpluses of about 1.46%, 1.92% and 2.51%, respectively. The angles serve as a sensitivity test only; no observational epoch was chosen here, and no full vector history was computed. The Moon's present spin is very small, but its early spin was larger, and the two must be kept apart.

Sensitivity to structure: the surplus depends on how uniform the parent was. At fixed figure, varying k gives a threshold of about 0.397 against the scalar sum, and about 0.390–0.394 against the vector range: when k is below this value, the parent's budget would fall short of today's. Today's differentiated Earth, with its dense core, has k of about 0.33, well below the threshold. The route therefore requires that the parent was still at an early stage of sorting when the transfer began; this requirement is listed in §D.3. This is a sensitivity condition, not a stratified Maclaurin family of solutions. The positive surplus shows that this test finds no angular-momentum deficit, but matching the rate of loss to the Sun requires a further calculation.

A.6.4 Sensitivity to period and expansion

A shorter initial period means more angular momentum. At the same time, a more expanded parent has a larger moment of inertia, and holds the same angular momentum at a longer day. The admissible initial period therefore depends on the adopted expansion.

The working window: in the reference model, the window 4.90–5.54 hours refers to the initial period and to the adopted budget and loss conditions. At 5.54 hours the parent holds exactly today's angular momentum of the system; at 4.90 hours it holds about 15% more — the adopted limit for loss to the outside. This ceiling must not be applied to the final period after the transfer of angular momentum, which is longer by design (§A.4). Appendix B retains the original sensitivity table, distinguishing between an adopted expansion and a thermal calculation not yet performed.

Figure 5. The admissible parameter region. The solid line marks $L_{\mathrm{initial}} = L_{\mathrm{present}}$ (table of Appendix B); above it the primordial budget falls below the present one. The dashed line marks the adopted external-loss limit, 15%. The point is the reference value.

Map of admissible starting states: Figure 5 is read as a map of starting states. The horizontal axis shows how much the hot parent was expanded beyond its cold radius; the vertical axis is the length of its day. A starting state above the solid line holds less angular momentum than today's system, and is therefore excluded. A state below the dashed line holds more than 15% extra, more than the external loss can remove. The band between them is admissible, and it rises to the right because a larger parent holds the same angular momentum at a longer day. The reference state — 5.5 hours with an expansion of 120 km — lies inside the band, just below the ceiling.

A.6.5 Comparison of equilibrium figures

The classical objection to fission can now be placed exactly. In the classical route the whole Earth must spin so fast that it becomes unstable and divides.

The sequence of shapes of a spinning fluid: a uniform spinning fluid passes through a known sequence of shapes as its angular momentum grows. At first it is a Maclaurin spheroid (§A.6.1) — flattened at the poles but round at the equator. At a definite flattening a second family branches off: the Jacobi ellipsoids, whose equator is no longer a circle but an ellipse, so the body becomes elongated like a rugby ball turning about a short axis. Further along the Jacobi sequence comes a pear-shaped figure, from which classical fission was expected to proceed. Each of these points requires a definite spin and a definite angular momentum.

The bifurcation point: at the classical bifurcation point from Maclaurin to Jacobi, with $e \simeq 0.81267$ and $\omega^2 / (\pi G \rho) \simeq 0.37423$, one obtains for the adopted mass and volume $T \simeq 2.73$ hours, $a \simeq 7,802$ km and an angular momentum of about 9.42×10^{34}, a ratio of about 2.74 to the present sum [1]. Here e is the eccentricity of the meridian section (zero for a sphere, approaching one for a very flat body), and $\omega^2 / (\pi G \rho)$ measures the spin against the body's own gravity. This means that merely to reach the first stage of the classical route, the Earth would have needed a day shorter than three hours and almost three times the angular momentum the Earth–Moon system holds.

Reference values in the literature: this is a comparison with a defined configuration on this classical route. The value 4.08 of the original reference calculation refers to a different point on the Maclaurin sequence — the maximum-Ω point, not the Jacobi bifurcation — and is not presented as a universal bound; it is the source of the familiar statement that fission needs about four times the available angular momentum. The value 3.5 for the pear-shaped point is not adopted, because verification of the table has not been completed.

The source of the difference: the heart of the difference is the place where the speed condition is evaluated. The classical route requires the parent's surface to reach orbital speed (§A.1.2); this requirement pushes the parent towards break-up, and break-up costs several times the available angular momentum. The LEF route never uses that condition: the material is carried out in corotation, and the speed match is made at the corotation radius (§A.5), where a parent turning once every 5.5 hours already supplies orbital speed. When the requirement is evaluated there, it falls to the budget the parent actually holds. The comparison is between two routes to the same product: the classical route needs 2.74 times today's budget; this route needs 1.007.

A.6.6 The vector correction for release off the equator

So far the material was assumed to leave in the equatorial plane. Release off the equator requires the spin vector and the orbital angular momentum to be summed in the same frame of reference, because the orbit of a body released at a given latitude is tilted against the spin axis. Let $S = I\omega$ be the spin angular momentum and $H = \mu_{\text{red}} \omega r^2 \cos \lambda$ the orbital angular momentum of the released body, where λ is the angle of release from the equatorial plane. For the geometry defined in Appendix D:

$$L_0^2 = S^2 + H^2 + 2SH\cos\lambda.$$

This is the same rule for adding two arrows at an angle as in §A.6.3: the initial budget must equal the vector sum of the spin and the tilted orbit.

Sensitivity analysis: in the example $\lambda = 25^\circ$ relative to the final axis, with a homogeneous structure, the coupled two-mass calculation gives a final period of 6.220 hours, a reference distance of about 17,240 km and an orbital angular momentum of about 15.83% of the budget. The 25° value serves as a sensitivity test of how much an off-equator release changes the budget; the adopted release latitude, the centre of the contact area, is 22.5° (Paper 4). This is a sensitivity correction to the budget, not a new reference value. The derivation of the inclination, the eccentricity and the perigee, and the evolution after release, are handed to Paper 4.

A.6.7 The budgetary decision condition

The test requires consistency between the initial state, the final state and the external transfer:

$$\mathbf{L}_i = \mathbf{L}_f + \Delta\mathbf{L}_{\text{external}}.$$

The equation is conservation of angular momentum written as an account: the angular momentum the system started with equals what it holds now, plus what was removed from it to the outside (for the Earth–Moon system, chiefly by the Sun's tide). Closing the numerical balance proves consistency between the defined states. Failure to find a match within the supported ranges of structure and external transfers would rule out this budgetary scenario. The match does not establish how the body mechanically reaches the final state; that is the subject of Paper 3.

Summary of the budget: the parent with a 5.5-hour day holds slightly more angular momentum than today's system; the transfer costs about one sixth of it; and the classical route would have needed 2.74 times as much.

A.7 The Roche limit

A body that reaches the corotation radius must also remain whole there. Close to a massive parent, the parent's gravity pulls harder on the near side of the body than on the far side; this difference, the tidal force, tends to stretch the body and pull it apart. The Roche limit is the distance inside which the tidal stretching exceeds the body's own gravity, which holds it together. Inside this limit, a body held by its own gravity alone cannot remain whole.

A.7.1 Two reference limits

For the adopted lunar density $\rho_s = 2,600\ \mathrm{kg\,m^{-3}}$ and the parent mass after departure, two Roche limits are obtained:

$$d_{\text{fluid}} = 2.44\left(\frac{3M_\oplus}{4\pi\rho_s}\right)^{1/3} \simeq 19,971\ \mathrm{km},$$

$$d_{\mathrm{rigid}} = 1.26 \left(\frac{3M_{\oplus}}{4\pi\rho_{s}} \right)^{1/3} \simeq 10,314\,\mathrm{km}.$$

(Note: an earlier version gave 10,313 km; the two values are the same within rounding, and the series uses 10,314 km. In Figures 1, 6, 8 and 9 the fluid limit is printed as 19,972 km, the same value within rounding.) The bracketed term represents the hypothetical radius of a sphere of the Earth's mass if it had the satellite's mass density; the limit is a fixed multiple of it. The two formulas differ only in this multiple, and the reason is physical. **The fluid approximation** describes a body that can deform continuously: under the tidal field it lengthens toward the parent, its near and far ends move farther apart, the tidal difference across it grows, and it gives way sooner — so the fluid limit lies farther out. **The so-called rigid approximation** neglects the change of shape: the body keeps its round form, the tidal difference across it stays smaller, and it can survive closer to the parent. This approximation contains no crustal thickness or material strength [1]; it represents a body held together by its own gravity alone that merely does not change its shape. The base radius and the two coupled radii lie in the range between these two limits: inside the fluid limit and outside the rigid one. The presence of an outer crust alone therefore does not establish the body's survival; survival depends on the body behaving as rigid over the time-scale of its stay in this region.

Sensitivity to density. The Roche limits move inward as the satellite density increases, because a denser body is held together more strongly by its own gravity. A density of about 4,700 kg per cubic metre would move the fluid limit inward to the release distance, about 16,400 km; for the rigid body at release the rigid limit applies (§B.7). This is a requirement computed in the Roche approximation, not evidence that the body reached that density, nor a solution to all the stability requirements.

A.7.2 What the comparison of accelerations teaches

The basic assessment of the satellite's survival is made by a direct comparison between the parent's gravitational acceleration gradient and the self-gravity acceleration at the satellite's surface:

$$\Delta a \simeq \frac{2GM_{\oplus}R_{s}}{r^{3}}, \qquad g_{s} = \frac{Gm}{R_{s}^{2}}.$$

The first quantity, Δa, is the difference between the parent's pull at the centre of the orbiting body and its pull at the body's surface, which is closer by a distance R_{s}; this gradient falls steeply, as the cube of the distance r. The second, g_{s}, is the body's own surface gravity, which holds its material in. When the first is smaller than the second, the body's own gravity dominates and the survival condition is met. Substituting the reference data of Appendix E at the initial corotation radius gives a ratio below unity (about 0.277 at 15,818 km), that is, the body's own gravity clearly prevails. This ratio serves as a working measure of loading, but the change of shape and the rotation are not included in it. The static Roche limit provides a basis for comparison; the response in an eccentric orbit may also depend on the deformation history [9].

A.7.3 What remains conditional

Reaching the corotation radius closes only the velocity gap; it is not a strength threshold and not a guarantee of survival. The properties of the body and of the connecting element (the pear) are examined in Paper 3, and the orbit after separation in Paper 4. Paper 2 adopts no binding residence time in the Roche domain, and no universal viscosity threshold. The survival assessment should refer to the closest distance to the parent in the chosen orbit, not to the release distance alone. Dividing the tests among the papers does not remove the requirement of survival.

Figure 6. The range of Roche arcs: rigid (solid) and fluid (dashed) limits for densities 1,300–3,344 kg/m^3, with the pear and the Moon at release.

Figure 6 maps the two families of limits for this range of densities and places the Moon at release against them. The rigid limits, for the whole range of densities, lie well inside the Moon's position; the fluid limits lie beyond it. The Moon at release is therefore safe as a rigid body, and its flattening and rounding were completed earlier, while it was still attached to the parent (§B.7; Paper 7).

A.8 The direction of tidal evolution

The last step of Part A examines the direction of the Moon's orbital motion immediately after separation: does the Moon move away from the Earth, or fall back onto the parent? The direction is set by the body's position relative to the corotation line.

A.8.1 The direction of angular-momentum transfer

For a direct, circular, equatorial orbit, we compare the parent's angular spin rate with the mean motion — the angular velocity of the orbiting body around the parent, defined by Kepler's law:

$$n = \sqrt{\frac{GM}{r^3}}.$$

The orbiting body raises a tidal bulge on the parent, and two states are defined by the ratio of the two speeds. **Orbital recession.** When $\omega > n$, the parent turns faster than the body orbits, and its rotation carries the bulge ahead of the body; the bulge pulls the body forward, gives it angular momentum, and the body climbs outward while the parent's spin slows. In the language of the tidal equations, the dissipative tidal contribution in the parent transfers angular momentum from its spin to the orbit. **Orbital approach.** When $\omega < n$, the body outruns the bulge, the bulge lags behind it and pulls it back, and the direction reverses: the body loses angular momentum and spirals inward. **Exact synchronism.** At exact circular synchronism the relevant contribution vanishes: the bulge stays directly beneath the body, and there is no forward or backward pull. Since n falls with distance and ω does not, $\omega > n$ holds outside the corotation radius and $\omega < n$ inside it. The corotation line is thus a tidal watershed: a body released just beyond it is pushed outward from the moment of release. Generalisation to an orbit with inclination and eccentricity requires the appropriate tidal equations [2,3].

Figure 7. The synchronous radius as a tidal watershed. Phobos orbits inside it and spirals inward; Deimos orbits outside and recedes. Schematic spacing, not to scale.

Figure 7 illustrates the watershed in the present Solar System. Mars has two small moons on opposite sides of its synchronous radius: Phobos, inside, is drawn inward; Deimos, outside, moves away. The Earth–Moon case at release is the same rule, applied to a Moon placed just beyond the line.

Summary of the threshold conditions of Part A. With Part A complete, the problems are fully defined: a velocity gap that closes at the corotation radius; a pear that reaches that radius; a day at release set by the offset between two opposing processes — the lengthening pear, which slows the spin, and the sinking core, which speeds it up; an angular-momentum budget that must close against today's system; a Roche test the body must pass; and a tidal watershed that decides the direction of its later motion. Part B answers all of them with one calculation.

PART B

How the problems are solved

> **THE PROBLEM AND HOW IT IS TESTED**
>
> One calculation answers the problems of Part A. Section B.1 sets out that calculation, from the input data to the state of the Earth and the Moon at the moment of release. Each of the following sections (B.2–B.8) takes one problem of Part A, in the same order, and shows in two or three sentences how the calculation answers it.

Part A defined the conditions; this Part presents one consistent, closed set of numbers that meets all of them at once. This carries weight, because each condition alone can usually be met by tuning one parameter, while meeting all the conditions together, with the same Earth, the same Moon and the same spin, is a much stronger test of the route.

B.1 The calculation

Basic data. Initial Earth day $T_0 = 5.5$ h (§A.2); a homogeneous, thermally expanded parent, with a volumetric radius of 6,517 km and a mean density of 5,214 kg m^{-3} (equatorial radius 6,721 km, polar radius 6,128 km); the transferred body (the Moon): one lunar mass, $m = 7.35 \times 10^{22}$ kg, at a density of 2,600 kg m^{-3}. "Homogeneous" means that the parent is defined at the start of the process as uniform in density — a calculation input for a parent still at an early stage of sorting; "expanded" means that it is hot and therefore somewhat larger than a cold Earth of the same mass

(§A.2). The equatorial and polar radii differ because a body completing one turn in 5.5 hours is hydrostatically flattened: it bulges at the equator and is compressed at the poles.

From Part A to the release state, step by step. The evolution of the system toward the release state follows from the processes of Part A in five successive steps, which balance slowing (the lengthening pear) against speeding up (iron sinking into the core). The table below collects them; its rows are read from top to bottom, and the tint shows the direction of each step: red for a step that slows the Earth and moves the corotation line outward, blue for a step that speeds it up and moves the line inward.

1. *The starting point.* For a parent with a 5.5-hour day, the corotation line lies at 15,818 km (§A.2.1). The corotation line is the distance at which a body turning together with the Earth already moves at the speed of a free circular orbit.

2. *The lengthening pear slows the Earth.* As the pear lengthens, material moves outward while still turning with the Earth, and it carries angular momentum with it. Angular momentum is the "quantity of rotation" of a body — its mass times its speed times its distance from the axis; with no external torque, its total is conserved. The angular momentum that leaves with the material is taken from the Earth's spin, so the Earth turns more slowly: alone, this step would lengthen the day from 5.50 to 6.44 hours and move the line out to 17,569 km (§A.4).

3. *The centralising core speeds up the Earth.* At the same time iron sinks toward the centre. This lowers the Earth's moment of inertia — the measure of how far its mass lies from the axis — and, like a skater on the ice drawing in her arms, the Earth turns faster. With a core twice as dense as the mantle above it ($D = 2$), this step shortens the day by 0.67 hours.

4. *The net result.* The two steps act together and partly offset each other: the day at release is 5.77 hours, and the corotation line lies at 16,331 km when the Moon is treated as a test particle, or at 16,399 km when the Moon's own mass is included in the two-body formula below. We use the two-body value, rounded to about 16,400 km.

5. *Release.* The Moon begins to separate when its centre of mass reaches this line. It is carried there by the rotation, not thrown: at that distance, the tangential speed it already has as part of the turning system is the speed of a free orbit, so no extra push is needed (§B.2).

Step	Length of the Earth day	Radius of the corotation line	Direction of effect
1. Starting point (5.5-h parent, §A.2.1)	5.50 h	15,818 km	reference state
2. The pear lengthens (alone)	6.44 h	17,569 km	slowing (mass and momentum carried out), line outward
3. The core centralises ($D = 2$)	-0.67 h	—	speeding up (sinking iron), line inward

Step	Length of the Earth day	Radius of the corotation line	Direction of effect
4. Combined solution (test particle)	5.77 h	16,331 km	net: modest slowing (partial offset)
4a. Combined solution (two-body)	5.77 h	16,399 km $\simeq$ 16,400 km	adopted value
5. Moon's centre of mass on the line	5.77 h	$\sim$16,400 km	release (tangential = orbital speed)

The quantitative content of these steps is the following.

1. The Earth day and spin rate at release. Combining the offset between the lengthening pear and the centralising core (§A.4) fixes, with $D = 2$, the length of the day and the angular spin rate at the moment of separation:

$$T = 5.77\,\mathrm{h}, \qquad \Omega = \frac{2\pi}{T} = 3.025 \times 10^{-4}\,\mathrm{s}^{-1}.$$

Here T is the length of the day and Ω the angular spin rate — the angle in radians through which the Earth turns each second. The day is only about a quarter of an hour longer than the initial 5.5 hours, because the speed-up caused by the sinking iron cancels most of the slowing caused by the pear.

2. The release point and the extent of the Moon along the line. The Moon begins to separate when its centre of mass reaches the corotation line:

$$r = R_{\mathrm{sync}} = \left(\frac{G\,(\,M_{\oplus} + m\,)}{\Omega^2} \right)^{1/3} \simeq 16,400\,\mathrm{km}.$$

This is the corotation formula of §A.5.1, written for a two-body system: G is the gravitational constant, $M_{\oplus}$ the Earth's mass and m the Moon's mass. In words: the formula gives the one distance at which a circular orbit takes exactly one turn of the Earth. Including m in the formula is what turns 16,331 km into 16,399 km. This distance is about 2.6 times today's Earth radius. With $R_m = \left(\frac{3m}{4\pi\rho} \right)^{1/3} = 1,890$ km, the near side of the Moon lies at 14,509 km and the far side at 18,289 km from the Earth's centre. R_m is the radius of a sphere of one lunar mass at a density of 2,600 kg m^{-3}; subtracting it from and adding it to the 16,400 km of the centre gives the extent of the Moon along the line between the Earth and the Moon.

3. The equilibrium shape of the Earth at release (Maclaurin, homogeneous):

$$\frac{\Omega^2}{\pi G \rho_{\oplus}} = \frac{2\sqrt{1-e^2}\,\left(\,3 - 2e^2\,\right)}{e^3} \arcsin e - \frac{6\left(\,1-e^2\,\right)}{e^2} \quad \Rightarrow \quad a = 6,701\,\mathrm{km},\ c = 6,165\,\mathrm{km}.$$

A Maclaurin spheroid is the equilibrium shape of a uniform fluid body spinning at a constant rate: a sphere flattened at the poles. The equation links the spin frequency (the left side, where $\rho_{\oplus}$ is the Earth's mean density) to the eccentricity of the meridian section e, a number that measures the flattening (0 for a sphere, approaching 1 for a flat disc). Solving it for a spin of 5.77 hours gives the equatorial radius a and the polar radius c. The Earth at release thus keeps a clear flattening, though

slightly less than that of the 5.5-hour parent (6,721 and 6,128 km), because it now turns a little more slowly. The gap between the Earth's equatorial surface and the Moon is about 7,800 km: the Moon's near side at 14,509 km less the equatorial radius of 6,701 km. This is the distance the pear must bridge (§A.3).

4. The angular-momentum budget and its division at the moment of separation.

$$L_0 = k_0 M a_0^2 \Omega_0 = 3.466 \times 10^{34}, \qquad L = I_\oplus \Omega + \mu\, \Omega\, r^2.$$

The first expression is the initial budget of the parent (§A.6.1, in kg m^2 s^{-1}): the moment-of-inertia coefficient k_0 (0.4 for a uniform body), the mass, the square of the equatorial radius and the initial spin. The second expression divides the same budget at release into two components: the Earth's own spin, $I_\oplus \Omega$, where $I_\oplus$ is its moment of inertia, and the Moon's orbit, $\mu \Omega r^2$, where $\mu = M_\oplus m / \left(M_\oplus + m \right)$ is the reduced mass of the pair and r the release distance. At release the Moon's orbit holds 5.9×10^{33} (17% of the budget) and the Earth's spin 2.88×10^{34}; the total angular momentum of the system is 1.007 times today's value (an earlier version gave 1.008, from today's total rounded to 3.44×10^{34}; with 3.441×10^{34} of §A.6.3 it is 1.007, the value used throughout the paper). The 5.9×10^{33} is defined as the orbital angular momentum of the pair about their common centre of mass, $\mu \Omega r^2$, with the reduced mass $\mu = 7.26 \times 10^{22}$ kg and $r = 16,399$ km on the equator. **Methodological note:** counting the full lunar mass m about the Earth's centre, as Paper 3 does, gives for the same state 6.0×10^{33}; for release at a latitude of 22.5° ($r = 17,287$ km, Paper 4), where the Moon turns at a distance of $r \cos 22.5^\circ = 15,971$ km from the spin axis, the same count gives 5.7×10^{33} (Paper 3). The 17% is the price of the transfer: the share of the original rotation that was subtracted from the Earth's spin and moved into the Moon's orbit. The total, 1.007 times today's value, is slightly above the present value, and this is the sign required, since solar tides have removed about 1% of the system's angular momentum since then (§A.2).

> **RESULT**
>
> Earth day at release **5.77 h**; Moon centre at ∼**16,400 km**; angular momentum **1.007** times today's.

Figure 8 draws this state to scale. It shows how close the Moon still is to the Earth at release: a flattened Earth, and a gap of about 7,800 km between it and the near side of the Moon, which the pear of §A.3 bridges in full.

Figure 8. The Moon at release, to scale: flattened Earth at 5.77 h, Moon centre on the corotation line.

We now have a complete, closed release state: the Earth's day and shape, the Moon's position and size, and the division of the angular momentum between them. The following sections check this state, section by section, against each of the problems of Part A.

B.2 The velocity gap

The problem of §A.1 was that material released from the equatorial surface of an Earth with a 5.5-hour day moves at about 2,133 m s^{-1}, far below the 7,701 m s^{-1} of a circular orbit — a gap of about 5,568 m s^{-1}. The gap closes with distance, because the tangential speed given by the rotation grows linearly with distance, while the orbital speed falls. On the corotation line the rotation gives the Moon exactly the orbital speed,

$$v = \Omega r = \sqrt{\frac{G\,(\,M_\oplus + m\,)}{r}} \approx 4,960\ \mathrm{m\,s^{-1}},$$

so no gap remains (§A.1, §A.5). The left side is the speed of a point turning with the Earth at distance r; the right side is the speed of a free circular orbit at the same distance; on the line the two are equal. The value is a little below the 5,020 m s^{-1} of the 5.5-hour reference state (§A.2.1), because the Earth, turning slightly more slowly, places the line farther out, where orbital speeds are lower. The mass is released into an independent orbit by virtue of the speed it already has from the rotation, with no need for an external push. The everyday picture is a child on a roundabout: the farther from the centre she sits, the faster she moves, without anyone pushing her.

B.3 The pear

The pear bridges a gap of about 7,800 km and lies entirely inside the corotation line; the Moon is released when its centre of mass reaches the line (§A.3; the release mechanism is detailed in Paper 3, *LEF Mechanism in Practice*). Because the pear lies inside the line, the net force in the turning frame is directed inward, toward the Earth (§B.5): the pear's material stays bound to the parent, while the Moon, whose centre of mass is on the line, is free to separate. This is why the Moon leaves as a whole unit, and the connecting element does not go with it.

B.4 The offset

The calculation solves the two opposing processes together and fixes the Earth day at 5.77 hours and the corotation line at 16,331 km (§A.4). The 16,331 km is the test-particle value of step 4 in §B.1; including the Moon's own mass in the two-body solution gives 16,399 km, about 16,400 km, used for the release point. A calculation based on the lengthening pear alone would give 6.44 hours and 17,569 km, and one based on the centralising core alone would give too fast an Earth; only the two processes together describe the Earth of that time.

B.5 Corotation

The equation of net acceleration. The Moon's centre of mass reaches the corotation line at about 16,400 km; beyond it corotation pulls the Moon outward, and it is released:

$$a_{\mathrm{net}} = \Omega^2 r - \frac{GM_\oplus}{r^2}.$$

This is the net acceleration felt by material that turns with the Earth at distance r, as seen in the turning frame. The first term is the outward (centrifugal) acceleration of the rotation, which grows with distance; the second is the Earth's gravity, which falls with distance. This equation fixes the

direction of the forces in the system: inside the corotation line the Earth's gravity dominates and a_{net} points inward, which keeps the pear joined to the Earth; on the line the two balance; beyond the line the centrifugal acceleration prevails and the material is drawn outward. As soon as the Moon's centre of mass crosses the point of balance, the body as a whole is subject to an outward pull that carries it away from the Earth, and it separates.

Figure 9 enlarges the moment at which separation begins. The line is drawn through the Moon's centre; the near half of the Moon still lies inside it and the far half outside.

Figure 9. The corotation line passes through the Moon's centre at the start of separation.

B.6 The angular momentum

The angular-momentum budget. The total angular momentum of the system stands at 1.007 times today's value. At release, the spin angular momentum remaining to the Earth requires a moment-of-inertia coefficient of $k = 0.355$; a two-layer structure with $q = 0.55$ and $D = 2$ gives

$$k = \frac{2}{5}\,\frac{\left(1 - q^5\right) + Dq^5}{\left(1 - q^3\right) + Dq^3} = 0.360.$$

The coefficient k describes how the mass is distributed: 0.4 for a uniform sphere, and lower when the mass is concentrated toward the centre (today's Earth has about 0.33). The value 0.355 is what is required of the Earth at release if a spin of 5.77 hours is to carry the remaining 2.88×10^{34}.

Fit to the two-layer model. The formula computes k for an Earth of two layers: an inner layer reaching to a fraction q of the radius, with a density D times that of the outer layer. With $q = 0.55$ and $D = 2$ it gives 0.360, close to the required 0.355. The structural value produced by the centralisation and sinking of iron thus agrees with the angular momentum the release state demands.

No high budget is needed. The route closes its budget within what the Earth–Moon system holds today, plus the roughly 1% that solar tides have removed since then. There is thus no need for a giant impact from outside, or for a later mechanism to remove surplus angular momentum, which characterise the high-angular-momentum impact models that start at about 2–3 times today's value.

B.7 The Roche limit

The Roche limit is the distance inside which the tidal pull of a large body — the difference between its gravity on the near side and on the far side of a smaller body — exceeds the smaller body's own gravity and can pull it apart. Two classical forms bound it:

$$d_{\text{rigid}} = 1.26\,R_\oplus \left(\frac{\rho_\oplus}{\rho}\right)^{1/3}, \qquad d_{\text{fluid}} = 2.44\,R_\oplus \left(\frac{\rho_\oplus}{\rho}\right)^{1/3}.$$

The rigid form applies to a body that keeps its shape; the fluid form to a body that deforms freely and is therefore stretched further by the tide. In both, ρ is the density of the small body: the denser it is, the closer it can come.

Position of the release point. The Moon's centre of mass at release lies at 16,400 km, well outside the rigid limit (9,484–12,994 km for 1,300–3,344 kg m^{-3}), but inside the fluid limit for a body of 2,600 kg m^{-3} (19,971 km, §A.7.1). Which of the two limits applies is therefore decided by the mechanical behaviour of the Moon at release: whether it is rigid.

Mechanical state at release. The rounding was completed before release; the body is rigid and round. The flattening and the later rounding of the lunar body took place earlier, while it was still attached to the parent (Paper 7, *The Poles*); what leaves the line is a body with a rigid outer shell.

Restraint of tidal deformation. The rigidity of the body reduces the tidal response by the factor

$$\frac{1}{1+\dfrac{19\mu}{2\rho g R}},$$

so that for $\mu \geq 10$ GPa the axis ratio is at most 1.04. The equation links the rigidity (shear modulus) of the body, μ, its surface gravity, g, and its radius, R; $\rho g R$ is the pressure of the body's own weight. The factor says that a body whose rigidity is large compared with the pressure of its own weight deforms only a small fraction of what a fluid body would. An axis ratio of at most 1.04 means that the longest axis of the Moon exceeds the shortest by no more than 4%: the body remains round.

B.8 The direction of tidal evolution

The mechanism of recession. The Moon is released on the corotation line, and the tide transfers angular momentum from the Earth's spin to the Moon's orbit: it recedes (§A.8; the orbit is the subject of Paper 4, *The Orbit*). Exactly on the line the tidal effect is neutral, because the Earth turns once in the same time as the Moon completes an orbit. Separation itself takes place when the Moon's centre of mass has moved about 210 km beyond the line, a stage at which a little more than half of the Moon's volume has crossed it (the margin $\Delta r \simeq 2d_0$ of §D.3). In this range the Earth spins faster than the Moon orbits; the tidal bulge raised on the Earth runs ahead of the Moon, pulls it forward and moves it outward in its orbit. This is the side of the watershed on which Deimos lies today (Figure 7).

How far beyond the line. The distance beyond the line is small compared with the release radius, but it sets the shape of the first orbit. On the line the Moon carries exactly the speed of a circular orbit; beyond it, a little more, so the release point becomes the nearest point of a slightly stretched orbit, with an eccentricity e_0 that grows with the distance. The table gives the sensitivity values for release at the release latitude, 22.5°, where the corotation line lies at 17,287 km, with a 5.77-hour day; the initial orbit comes from the formulas of Paper 4 (Appendix A), and the final angle from the reduced tan-law of Paper 4 (Part D). "Moon's volume crossed" is the share of the Moon's volume lying beyond the line at release.

Distance beyond the line (km)	Moon's volume crossed	Initial semi-major axis a_0 (km)	Initial eccentricity e_0	Final angle (reduced law)
0 (release on the line)	50%	17,287	0	5.020°
50	52%	17,489	0.009	5.049°
100	54%	17,696	0.017	5.079°
150	56%	17,907	0.026	5.109°
≈ 210 (adopted state)	**58%**	**18,163**	**0.037**	**5.145° (observed value)**
300	62%	18,571	0.053	5.202°
500	69%	19,531	0.089	5.334°
1,000	86%	22,404	0.184	5.711°

Table B.8. Release at 22.5°: the initial orbit and the final angle of the reduced law as the release moves beyond the corotation line (Paper 3, §C.5a). The navy deepens with the final angle; green marks the observed 5.145°, which is the adopted release.

Fit to today's values. Release exactly on the line gives a circular orbit, and the tide carries a circle outward as a circle; the Moon's present eccentricity therefore requires a release a little after its centre of mass has crossed the line. The adopted release margin, about 210 km, returns the observed orbital inclination by the reduced tan-law, and Paper 3 (§C.5a) shows that the same margin returns today's eccentricity, 0.0549, for a hot young Moon; the series adopts this release. Paper 3 gives the eccentricity table and the sensitivity to the latitude of the tip; the release state is fed as input to the full integration of Paper 4, which follows it through the whole orbital history.

What we now have: the release state of §B.1 meets each of the seven problems of Part A with one set of numbers.

PART C

Summary

C.1 What has been shown

The model of Part B answers all the problems of Part A within one set of numbers, without separate working assumptions for each problem. The table pairs each problem of Part A with the answer given to it in Part B. It is read row by row; each check mark means that the release state of §B.1 meets that problem without a separate assumption.

Problem	Solution
The velocity gap	the speed is supplied by the rotation; on the line the gap vanishes
The pear	bridges to the line; the Moon is released whole
The offset	5.77 h — balance between the lengthening pear and the sinking iron
Corotation	centre of mass on the line, $\sim$16,400 km
Angular momentum	1.007 times today's; consistent with the core structure
The Roche limit	rigid body, well beyond the rigid limit
Tidal evolution	recedes from the moment of release (tidal recession)

PART D

Claims, tests, requirements and novelty

This Part collects in one place what the paper claims, how each claim is tested, and what the route requires of its mechanism, so that these points are found here rather than scattered through the text.

REQUIREMENTS

- **Continuous offset.** Core centralisation $D \approx 2$ during the transfer. The organisation of the core is one continuous process: it began before the transfer, proceeds through all the stages of the pear's growth, and goes on after release. (Without the sinking iron the pear alone would slow the Earth to 6.44 h, §B.4.)

- **Rigidity.** A rigid body ($\mu \geq 10$ GPa) at release. (A fluid body at 16,400 km would lie inside the fluid Roche limit, §B.7.)
- **Clean separation.** A continuous pear from the Earth to a contact disc 3,000 km wide, with no neck remnant left attached to the Moon. This is the condition for the Moon to leave whole; it is examined in Paper 3.

D.1 Central claims and tests

The table lists the five central claims. The first column names each claim and states its status; the second, what it assumes and predicts; and the third, how it is tested.

Claim / status	Assumptions and quantitative prediction	Test and sensitivity
Corotation — an application of standard physics	Central field, equator, defined ω and M; $R \propto M^{1/3}T^{2/3}$	Reproduction of the two velocities and check of field corrections
Closure of the angular momentum — a consistency test of the scenario	$L_0 \simeq 3.466 \times 10^{34}$; release with the centre of mass on the corotation line, about 16,400 km ($D = 2$)	Reproduction of I, vector budget and domain of external transfers
Sensitivity to structure — a computed result	A change of I in the two-layer model changes T_f and R_f	Scan of D and q; check of shape and convergence
Non-equatorial angular momentum — a geometric correction	Definition of λ relative to one axis for all vectors	Vector sum and check of all angular-momentum components
Survival — an open requirement	Defined density, structure and orbit	Roche comparison and deformation solution in the follow-on papers

In plain words, the claims are these.

- *Corotation.* The claim is that at the corotation line the speed given by the rotation equals the orbital speed. The claim is tested by reproducing the speeds and by checking the corrections the scenario needs (the field of a flattened Earth, the size of the Moon, the pull between the two bodies) against that equality.
- *Closure of the angular momentum.* The claim is that the starting budget of 3.466×10^{34} suffices for a release at about 16,400 km. It is tested by matching the budget before release to the budget after it, through the structure and including the loss to solar tides.

- *Sensitivity to structure.* The claim is that a sinking core, that is, a change in the two-layer structure, changes the day and the line in the computed way (§A.4.1). The claim is tested by a full scan; the calculation settles on a solution for each structure scanned.
- *Non-equatorial angular momentum.* The claim is that a release off the equator changes the budget only by the computed vector correction (§A.6.6). The check is made by summing all vectors in one frame of reference.
- *Survival.* The claim is that the Moon survives the Earth's tide at release. This is an open requirement, and it is tested in the following papers.

These claims are not all independent observational predictions: some are physical identities, some are consistency tests for the closure of the numerical values, and some are requirements on the mechanism passed to the follow-on papers. The distinction is kept in the novelty map of Appendix F. An identity, such as the corotation formula, cannot be wrong if the data are right; a consistency test checks that the numbers close; a requirement is a condition the next papers must meet.

D.2 The novelty of the work

The physical foundations of the model — the corotation formula, Maclaurin figures, conservation of angular momentum and the Roche limits — are existing physics. The novelty lies in their joint application to a connected-transfer condition, with a consistent computation of the change in mass, shape and spin, and with a presentation of the sensitivity domains. "Connected transfer" means that the material stays joined to the Earth and turns with it until it reaches the line, instead of being launched from the surface. Replacing the surface release condition with this condition is what makes the classical objection to fission — an angular-momentum budget several times larger than required — not arise (§A.6.5). A full claim of historical priority requires the prior-work check of Appendix F; the success of the numerical reproduction does not prove priority.

D.3 What is assumed and required

The assumptions and requirements supporting the argument are collected here to avoid repeating them alongside each result. Each row gives the value adopted or required.

Adopted or required	Value
Initial rotation period	5.5 h; working window 4.90–5.54 h at $\Delta R = 120$ km
Thermal expansion	120 km; admissible 98–520 km

Adopted or required	Value
Parent structure at the start of transfer	At an early stage of sorting, $k \geq 0.397$ (calculation input); the organisation of the core is one continuous process that goes on through the transfer and after release
Release condition	Full corotation up to release; release beyond the corotation radius by $\Delta r \simeq 2d_0 \approx 210$ km ($d_0 \approx 105$ km)
Survival at release	Interior viscosity of order 10^{18}–10^{20} Pa s, or density above $\sim$4,700 kg/m^3, at which the fluid Roche limit reaches the release distance of 16,400 km (§A.7.1; an earlier version gave $\sim$3,800 kg/m^3, the value for the earlier pear-alone distance of 17,569 km)
The tidal watershed	Satellites outside the corotation radius recede, inside it fall
The budget comparison	Scalar as baseline (0.7%), vector as sensitivity (1.5–2.5%)

A few rows need clarification. **Thermal expansion and structure:** the *thermal expansion* is the amount by which the hot young Earth was larger than a cold Earth; the volumetric radius at the lower end of the admissible range (98 km of expansion) is set at 6,495 km. **Separation margin:** the *release condition* is defined as follows. A body released exactly on the corotation line is in a tidally neutral state; as soon as it lies beyond the line, the tide pushes it outward, so the Moon recedes from the start (Paper 4, §B.2). Separation places the centre of mass beyond the line by a margin $\Delta r \simeq 2d_0$, where d_0 is the thickness scale of the connection (Appendix E, row E47); the margin is about 210 km, $d_0 \approx 105$ km, a working point set by the release-distance search of Paper 3, §C.5a (an earlier version gave about 100 km). **Budget-comparison methodology:** the *budget comparison* is made in two ways: summing the spin and orbital angular momenta as plain magnitudes (scalar), which gives a surplus of 0.7%, and summing them as vectors tilted at an angle to each other, which gives 1.5–2.5% (an earlier version gave 1.5–2.6%; the largest of the three tested angles gives 2.51%); the scalar sum serves as the baseline, and the vector sum as a sensitivity test (§A.6.3).

Classification of the parameter rows: the first three rows of the table are adopted values, each with a window; the next two rows are requirements the route places on Paper 3; the last two rows are the checks that Paper 2 itself performs. None of the seven rows is hidden inside a result, and none of them is needed to fix the corotation radius, which follows from the parent's mass and spin rate alone.

PART E

Conclusions

A body of the Moon's mass can reach orbit from the Earth's own material, without a giant impact from outside. The fast spin of the parent supplies the tangential speed; the mass bridge (the pear) connects the Earth to the corotation line; the lengthening pear and the sinking of iron into the core together set a day of 5.77 hours at release; the Moon, as a rigid and round body, is released when its centre of mass reaches the corotation line at a radius of about 16,400 km; the total angular-momentum budget closes at 1.007 times today's budget, consistent with the loss to solar tides; and the Moon recedes from the moment of release.

The process comes together as a single chain: the Earth's rotation carries the material outward; the line at which the carried tangential speed equals the orbital speed sets where the Moon separates; the balance between the pear and the core sets where that line lies; and the parent's initial angular-momentum reservoir suffices, with a surplus of about 1% that the solar tide will remove later.

E.1 The earlier conclusions retained

The following conclusions of earlier versions of this paper remain valid and are kept here, with their original reference values.

1. **The release condition as decisive:** the definition of the place of separation determines the velocity test; direct release from the surface and transfer in corotation are different initial conditions. A surface release leaves a velocity gap (§A.1.2), while a corotating transfer leaves none (§B.2).

2. **Radius of the initial reference line:** the original reference value is about 15,818 km, for a 5.5-hour day and in the mass definition set for it. It is the line before the transfer began; the release value of about 16,400 km is the line after the pear and the core have acted (§B.1).

3. **Angular-momentum percentages in the budget:** the addition the material must gain is about 15.6% in the base calculation and about 16.5% in the coupled solution; about 17.7% is the final orbital angular momentum in the test-particle approximation, for the coupled state with the pear alone ($D = 1$, 6.44 hours, 17,569 km); in the release state with $D = 2$ the Moon's orbit holds about 17.0% (§B.1). The three percentages measure different quantities: the first two measure the angular momentum that the material must gain during the transfer (§A.6.2); the third measures the share of the budget held by the Moon's orbit at the end.

4. **Sensitivity:** the internal structure and the vector geometry affect the result, and were therefore included as sensitivity tests, not as measured past values.

5. **The requirements on the mechanism:** corotation supplies the speed and the budget. Mechanical detachment beyond the corotation line, sustained recession and survival in the Roche domain are the three conditions the route places on its mechanism; they are collected in §D.3.

Further work — openings for research

The calculations in this paper fix the inputs for further work that other research groups are equipped to undertake. Each item names a calculation this paper does not perform, and the paper of the series that will use its result.

1. **The stratified Maclaurin problem.** The solution of the coupled end state (§A.6.2) uses a uniform figure combined with a varying moment-of-inertia coefficient. A Clairaut–Darwin–Radau treatment of a parent in which iron is sinking will derive the corotation radius for the centralising core self-consistently (about 16,300–16,400 km at $D = 2$), and will supply Paper 4 with its starting orbit. The Clairaut–Darwin–Radau method is the standard way of computing the shape of a rotating body whose density increases inward; it would replace the uniform shape used here with one that follows the sinking iron.

2. **The dissipation of the early Earth.** The recession from about 2.6 Earth radii (16,400 km; 2.8 Earth radii in the earlier pear-alone state, with a 6.44-hour day) is set by k_2/Q and by its history; the present value is anomalous. Here k_2 measures how much the Earth deforms under the tide, and Q how much of the tidal energy it dissipates; their ratio sets the rate at which the Moon recedes. Ocean-resonance models run from the 5.77-hour day at release (rather than from the 6.4-hour day of the earlier, pear-alone state) would give the rate function that Paper 4 requires.

3. **The thermal expansion of the parent.** The 120 km adopted here (98–520 admissible) is a first estimate from magma-ocean depth and melt expansion; a full equation-of-state integration for a hot, volatile-rich early Earth would fix it, and with it the rotation window.

4. **The onset of recession.** A body released exactly on the corotation line is in a tidally neutral state: the Earth turns once in the time the body completes one orbit, and the tide neither pushes it out nor pulls it in. As soon as the body lies beyond the line, the tide pushes it outward, so the Moon recedes from the start (§B.8; Paper 4, §B.2). Separation leaves the centre of mass beyond the line by the release margin $\Delta r \simeq 2d_0$, about 210 km (§D.3, §E.1). A tidal integration with a frequency-dependent rheology in both bodies, starting from the release state of Paper 3, would fix the range of the margin and the rate of the first recession. “Frequency-dependent rheology” means that the deformation response of each body to the tide depends on how fast the tidal field changes.

5. **Survival inside the fluid Roche limit.** The order-of-magnitude strain estimate of the appendix “Estimate of acceleration and strain” is not a solution of the deformation field. A full visco-elastic calculation of the rigid body of 1,890 km, staying at about 16,400 km from the Earth over 10^1–10^3 years, would replace it. A visco-elastic calculation treats the body as elastic over short times and slowly flowing over long ones, which is how a hot body with a rigid outer shell behaves.

6. **The classical route.** The factor 2.74 at the Jacobi bifurcation is computed; the pear-shaped point along the Jacobi sequence requires the tabulated values to be verified, and a modern computation of the sequence for the adopted mass and density would settle it. The Jacobi

sequence is the family of elongated, three-axis equilibrium shapes into which a fast-spinning fluid body passes beyond the Maclaurin shapes (§A.6.5).

Code availability

All numerical results of this paper can be reproduced with the scripts of the LEF reproduction package (Python 3, numpy, scipy), which prints the values of each section from the inputs stated in the text. The package is attached to this preprint as supplementary material (LEF_code_and_data.zip). An earlier version of the series and of the package is deposited at doi:10.5281/zenodo.22743615.

APPENDIX A

Derivation of corotation, definitions and constants

This appendix fixes the definitions on which every number of the paper rests. It supports the corotation radius of §A.2.1 and §A.5 (15,818 km for the 5.5-hour reference) and the release distance of §B.1. A reader who wants to check any result should begin here, because most apparent disagreements between reproductions come from a different choice of mass, distance or constant, not from a different physics.

The short full derivation is in §B.1. **Distinguishing the mass definitions.** M_{tot} is the total initial mass: the Earth and the lunar material together, as one body, before any part of it has left. $M_{\oplus}$ is the final parent mass: the Earth alone, after the lunar material has been assigned to the second body. Using one where the other is required counts the lunar mass twice, or not at all.

Defining distances in a two-body field. A radius measured from the parent's centre is not a radius measured from the centre of mass of the system; once a second body of real mass exists, the two bodies turn about their common centre of mass, which lies slightly away from the Earth's centre. In the two-body case the distance r is measured between the centres, and the gravitational parameter (the product of the gravitational constant and the mass that pulls) is $G\,(\,M_{\oplus}+m\,)$, because each body orbits under the pull of the other, and the relative motion is governed by the sum of the masses.

Status of the reference constants. The constants are the source inputs as reproduced, not a new fit to an ephemeris (a table of observed positions). Changing the definition of the mass therefore requires a new calculation, not a silent replacement of the label 15,818: that number belongs to one stated definition, and a different definition produces a different number.

APPENDIX B

The rotating geometry and the coupled angular-momentum budget

This appendix derives the shape and the angular momentum of the homogeneous, expanded parent before the transfer begins. It supports the initial budget $L_0 \simeq 3.466 \times 10^{34}$ kg m^2 s^{-1} of §A.6.1 and §B.1, the comparison with the present system in §A.6.3, and the admissible region of Figure 5 in §A.6.4. The steps are: fix the mass and the size; find the flattened shape a uniform body takes at a 5.5-hour spin; compute its moment of inertia; and multiply by the spin rate.

Appendix B.1 Derivation of the initial structure — the base text preserved

Step 1: fixing the mass values. The model adopts $G = 6.674 \times 10^{-11}$, $M_\oplus = 5.972 \times 10^{24}$ kg and $m = 7.35 \times 10^{22}$ kg. Before the transfer, the lunar material is still part of the Earth, so the parent carries both. In the calculation code the exact sum of the adopted values is kept:

$$M_{\mathrm{tot}} = M_\oplus + m = 6.0455 \times 10^{24}\ \mathrm{kg}.$$

The rounded source value is 6.045×10^{24}. The rounding difference explains small deviations between reproductions; it is not a physical difference.

Step 2: the size and the density. The reference radius derived from the Earth's mean density is about 6,397 km: this is the radius the whole initial mass would have at the density of the present Earth. The young Earth in the model is hot and expanded. An adopted expansion of 120 km gives a volumetric radius $R_{V,0} = 6,517$ km (the radius of a sphere with the same volume as the flattened body) and a mean density of about 5,214 kg per cubic metre.

Step 3: the Maclaurin flattening equation. A uniform fluid body that spins settles into a flattened figure, wider at the equator than through the poles. For a body of uniform density this equilibrium figure is a Maclaurin spheroid, and its shape is fixed by one equation [1]:

$$\mathcal{M}\left(e\right) = \frac{2\sqrt{1-e^2}}{e^3}\left(3-2e^2\right)\arcsin e - \frac{6\left(1-e^2\right)}{e^2}, \qquad \frac{\omega^2}{\pi G\rho} = \mathcal{M}\left(e\right).$$

In words: the left side, $\omega^2/\left(\pi G\rho\right)$, measures how fast the body spins compared with the strength of the self-gravity that holds it together (ω is the angular velocity, ρ the density). The right side, $\mathcal{M}\left(e\right)$, is the flattening that this balance requires, written through the eccentricity e of the meridian section — $e = 0$ is a sphere, and e grows as the body flattens. Given the spin and the density, the equation is solved for e.

Once e is known, the equatorial radius a and the polar radius c follow from the volumetric radius:

$$a = \frac{R_V}{\left(1 - e^2\right)^{1/6}}, \qquad c = R_V \left(1 - e^2\right)^{1/3}.$$

The two expressions keep the volume fixed: $a^2 c = R_V^3$, so the body flattens without gaining or losing material.

For a given spin below the maximum, the Maclaurin equation has two solutions: one slightly flattened and one strongly flattened. A body spun up gradually from a sphere follows the first. On the branch near the spherical shape one obtains $e_0 \simeq 0.4107$, $a_0 \simeq 6,721$ km and $c_0 \simeq 6,128$ km — an equatorial bulge in which the equator lies about 600 km further from the centre than the poles.

Step 4: the moment of inertia and the angular-momentum budget. The moment of inertia measures how hard a body is to set spinning, and depends on how far its mass lies from the axis. For a uniform spheroid turning about its short axis it equals $\frac{2}{5}$ of the mass times the square of the equatorial radius. The angular momentum is the moment of inertia times the angular velocity:

$$I_0 = \frac{2}{5} M_{\mathrm{tot}} a_0^2 \simeq 1.092 \times 10^{38}\ \mathrm{kg\,m^2},$$

$$L_{\mathrm{initial}} = I_0 \omega_0 \simeq 3.466 \times 10^{34}\ \mathrm{kg\,m^2\,s^{-1}}.$$

Here $\omega_0 = 2\pi/19,800\ \mathrm{s}^{-1}$, the angular velocity of a 5.5-hour day. The boxed number is the whole angular-momentum reservoir the transfer may draw on; §A.6 compares it with the present Earth–Moon system.

Uniform density is a calculation input. It represents a parent still at an early stage of sorting: iron may already have gathered in the core at the start, and the organisation of the core continued through the transfer and after release. A body that has not completed differentiation is not necessarily uniform; the effect of departures from uniform density is examined below in a sensitivity test (Appendix C and §A.4.1).

Appendix B.2 The original expansion scan

The 120 km expansion is an adopted value, so the question is how much the scalar budget model depends on it. The original sensitivity table answers this. Each row gives one expansion; the second column is the resulting volumetric radius; the third is the surplus of the initial budget over today's budget at a 5.5-hour day; the fourth is the longest initial day at which the budget still equals today's (the scalar ceiling). Read the table from top to bottom: as the expansion grows, the body becomes larger, its moment of inertia rises, and both the surplus and the ceiling grow with it.

Adopted expansion, km	Volumetric radius, km	Surplus at 5.5 hours	Scalar ceiling, hours
0	6,397	−3.3%	5.34
98	6,495	0.0%	5.50

Adopted expansion, km	Volumetric radius, km	Surplus at 5.5 hours	Scalar ceiling, hours
120	6,517	+0.7%	5.54
200	6,597	+3.5%	5.67
300	6,697	+7.0%	5.84
520	6,917	+15.0%	6.21 (an earlier version gave 6.22)

Limits of the admissible range (§D.3). Two rows mark the edges of the range. With no expansion the budget falls 3.3% short of today's, so an unexpanded parent spinning at 5.5 hours cannot supply the angular momentum of today's system. At 98 km the budget exactly equals today's; this is the smallest expansion that works. At 520 km the surplus reaches 15%, the adopted limit on what could later have been lost to the Sun. The range 98–520 km of §D.3 follows from these two rows.

Status of the numerical window. The table is kept from the source table as a sensitivity of the scalar model. The ceiling of 5.54 hours belongs to the reference state, not to every possible structure. The floor of 4.90 hours rests on an adopted external-loss cap of 15%; there is no independent derivation of that cap here. The range 4.90–5.54 hours is therefore not a proven observational constraint, but the window within which the adopted inputs remain consistent.

The 120 km is also an adopted parameter. Representing thermal expansion by this parameter does not replace an equation-of-state calculation (a calculation that follows how the density of the material changes with pressure and temperature, layer by layer). Establishing the expansion of a body under pressure requires integrating the structure and the density; a linear estimate based on layer depth and an expansion coefficient is not a closed independent basis. The further-work list names this calculation as an opening for research.

APPENDIX C

The differentiation calculation and the sensitivity tests

This appendix describes how the two-layer calculation underlying §A.4 and §A.4.1 was set up, and how far its numbers can be trusted. It supports the table of final periods and corotation radii against the density ratio D (for $D = 2$: 5.77 hours and 16,331 km), and the value $k = 0.360$ of §B.6. The

physical idea is the skater drawing in her arms: when iron sinks toward the centre, the moment of inertia falls, and with the angular momentum conserved, the spin speeds up.

Characterising the two-layer model. The body is divided into two layers: an inner one and an outer one. q is the ratio of the volumetric radius of the inner layer to that of the outer, and D is the ratio of their densities. The layers are incompressible and in shared rotation, so they turn together as one body. The volume of the final body is fixed in all scenarios; the initial state used to set L_0 remains homogeneous, so every scenario starts from the same budget and differs only in how the final body has arranged its mass.

Solving the equipotential surfaces. For each scenario a constant effective potential is solved on both boundaries: each boundary is required to be a surface of equal combined gravitational and centrifugal potential, which is the condition for a fluid layer to be at rest in the rotating frame. I is then computed directly from the density and the shape, not taken from a formula for a uniform body. There is no differentiation path in time and no equation of state here: the calculation compares end states, and does not follow the sinking of iron step by step.

The reliability of the results is examined by two convergence checks. **Freeing the shape of the surfaces:** allowing the layer boundaries to depart from an exact spheroidal shape (instead of forcing that shape on them) changes the radius by up to about 2.5 km in the $q = 0.55$ scan — a small deviation against a corotation radius of more than 15,000 km. **Numerical resolution:** changing the numerical step size changes the radii by only a few metres. The field residuals do not converge monotonically in every check, so the model makes no claim of metre-level physical precision. The results are reliable to the kilometre level, which is the working scale of the paper.

APPENDIX D

The vector correction to the budget and the hand-off to Paper 4

The main calculation releases the Moon in the equatorial plane. This appendix examines what changes when the release point lies off the equator, and it supports the budget equation of §A.6.6 and the sensitivity values quoted there. The point is geometric: off the equator, the orbital angular-momentum vector of the Moon and the spin angular-momentum vector of the Earth do not coincide on a single axis, and angular momenta that point in different directions add as arrows in space, not as plain numbers.

In a frame in which z is defined as the final spin axis, and λ is the latitude of the release point, measured relative to its equator, the total angular-momentum vector is defined as:

$$\mathbf{L}_{\mathrm{tot}} = \left(-H \sin\lambda, 0, S + H\cos\lambda \right), \quad H = \mu_{\mathrm{red}} \omega r^2 \cos\lambda.$$

where S is the spin angular momentum of the Earth along the axis. H is the orbital angular momentum of the Moon: μ_{red} is the reduced mass of the pair, ω the common angular velocity and $r\cos\lambda$ the distance of the release point from the spin axis — this distance alone fixes the tangential speed the rotation supplies. The orbital arrow is tilted by an angle λ from the spin axis, and so it splits into a component $H\cos\lambda$ parallel to the axis and a component $-H\sin\lambda$ perpendicular to it.

The length of the total vector follows from squaring the components and adding them:

$$L_0^2 = (H\sin\lambda)^2 + (S + H\cos\lambda)^2 = S^2 + H^2 + 2SH\cos\lambda.$$

This is the budget equation of §A.6.6: the initial budget must equal this length. When $\lambda = 0$ it reduces to the plain scalar sum $L_0 = S + H$ of the equatorial calculation. The angle between the spin axis and the total angular momentum is defined as:

$$\beta = \arctan\left(\frac{H\sin\lambda}{S + H\cos\lambda}\right).$$

This angle expresses how far the Earth's final spin axis leans from the angular-momentum axis of the whole system, and it is the starting point for the axis tilts that Paper 4 follows.

For the test case $q = 0.55$, $D = 1$, $\lambda = 25^\circ$ the solution of the equations gives a release distance of about 17,177 km and an angle of 3.85° in the test-particle approximation, or about 17,240 km and 3.84° in the two-mass solution. The 25° value serves only as a sensitivity test, chosen to show the size of the correction; the adopted release latitude is 22.5°, the centre of the contact area, and the release there is recomputed in Paper 4 (*The Orbit*). These are proposed sensitivity results, and they do not replace the reference values. The earlier numbers 17,115 and 4.48° were set apart in the log because they required a frame-consistency correction: they mixed vectors expressed relative to different axes. This kinematic equation describes the state at one defined moment, and does not compute the change of the axis as a function of time.

The full numerical documentation, including the orbital material intended for Paper 4: LEF_PaperI_Calculation2_Latitude25_HE.

APPENDIX E

Roche calculations and the survival assumptions

This appendix gathers the background to §A.7 and §B.7: two Roche limits and a first estimate of how strongly the Earth's tide loads the young Moon. The Roche limit is the distance inside which the difference between the Earth's pull on the near and far sides of a satellite (the tidal stretch) exceeds the self-gravity holding it together. A fluid body that deforms freely develops an elongation that

increases the stretch on itself, so its limit lies further out (coefficient 2.44); a body that keeps its shape has a limit further in (coefficient 1.26). The question this appendix answers is how large the tidal load is at the release distance, and what more is needed to turn that load into a survival result.

The coefficients 1.26 and 2.44 and the definitions of the limits were kept from the previous text. A focused verification of the source for the so-called rigid coefficient and of its domain of validity is defined as an open requirement to be completed before submission. The coefficient contains no strength law for a crust: the rigid limit describes a body that does not change shape, without saying what material strength makes this possible.

Estimate of acceleration and strain — a previous calculation text preserved

The estimate compares two accelerations. Δa is the tidal stretch: the difference between the Earth's pull at the centre of the body and its pull at the body's surface, along the line to the Earth. g_s is the self-gravity acceleration at the body's surface. For a body radius $R_s = 1,890$ km:

$$\Delta a \simeq \frac{2GM_\oplus R_s}{r^3}, \qquad g_s = \frac{Gm}{R_s^2}.$$

The stretch falls with the cube of the distance r: halving the distance raises it eightfold. At a radius of 15,818 km one obtains $\Delta a \simeq 0.381$, $g_s \simeq 1.373$ m/s² and a ratio of about 0.277. The ratio compares the tidal stretch with self-gravity: when it is below one, self-gravity wins, and at this distance the tidal load is a little over a quarter of the body's own gravity. This condition is necessary, but does not settle the question for a deforming body; the deformation of a hot body over the years it spends inside the fluid limit is the survival condition detailed in §D.3.

The next step converts the acceleration into a stress and the stress into a time. The stress estimate and the time for the deformation to develop are defined by:

$$\sigma_{\mathrm{tidal}} \sim \rho_s \Delta a R_s \sim 1.9 \times 10^9 \ \mathrm{Pa}, \qquad t_{\mathrm{strain}} \sim \frac{\varepsilon_* \eta}{\sigma_{\mathrm{tidal}}}.$$

The first expression represents the weight-like load of a column of the body's material, of density ρ_s and height R_s, under the acceleration Δa. The second describes how long a material of viscosity η (its resistance to flow, in pascal-seconds) takes to reach a strain ε_* — a fractional change of shape — under this stress: the stiffer the material, the longer the process.

Substituting $\varepsilon_* = 0.5$ (a change of shape by half) and $\eta = 10^{19}$ Pa s gives about 85 years (with $\rho_s = 2,600$ kg/m^3, $\Delta a \simeq 0.381$ m/s^2 and $R_s = 1,890$ km the stress is 1.87×10^9 Pa; an earlier version of this estimate gave 1.4×10^9 Pa and about 116 years, a value corresponding to a body density of about 1,950 kg/m^3). This is an estimate of strain under a constant representative stress, not a calculation of a disruption time. The rheology (how the material responds to load over time), the shape, the varying stress and the actual residence time must be examined.

This estimate is kept as a parametric dependence: it shows how the time changes with viscosity and with stress, and lets the reader insert other values. The residence time itself is derived from the orbital calculation and is not adopted here. The third calculation document collects the Roche, loading and orbit checks and the separation between a result and an open requirement: LEF_PaperI_Calculation3_Release_Survival_HE. The mechanical loading data are intended for Paper 3 (*LEF Mechanism in Practice*), and the orbital tables for Paper 4 (*The Orbit*).

APPENDIX F

Prior work, novelty and record of tests

This appendix maps, component by component, what the paper takes from earlier physics and what it adds, and it supports the novelty statement of §D.2. The table should be read row by row: the first column names a component of the argument, the second classifies it as standard physics, prior work or the paper's own application, and the third describes how the paper uses it. The pattern is deliberate: the basic laws are prior knowledge; what is new is where they are evaluated and how they are combined in one connected scenario.

Component	Classification	Treatment in the paper
Circular velocity and corotation	Standard physics [8]	Derivation and application, without a claim of priority for the law
Maclaurin and Jacobi	Prior physics [1]	Definition of the reference configuration and the domain of assumptions
Tides and the sign of angular-momentum transfer	Prior physics [2–4,6]	Presentation of the valid case and limitation of the generalisation
Two rotating layers	A method based on prior work [5]	The work's calculation and sensitivity analysis
Combination of mass, shape and budget in the connected scenario	Application of this work	Reproducible results; full priority open to examination
Roche in a dynamic body	Reference test with limitations [1,9]	No proof of survival from a single coefficient

Before submission a focused comparison with prior work on synchronous stability and equilibrium states is also required, to assess priority; in particular, the references to Counselman, Hut and

Grjebine & Marchal from the previous review record are not marked here as re-verified. These are open bibliographic tasks, not new results.

The editor's test is preserved: for every central claim its status must be identified, its assumptions and result defined, its sensitivity exposed, and a route to reproduction provided. The table of central claims in §D.1 (in the earlier text labelled "Table E.4"), together with the table of assumptions and requirements in §D.3, fills this structure; the absence of a survival solution is marked explicitly as open, and does not become a PASS.

References

[1] Chandrasekhar, S. (1969). *Ellipsoidal Figures of Equilibrium*. Yale University Press. Source from the existing text; exact page references for the sequence values and the Roche coefficients require completion before submission.

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[9] Yu, H., Lau, S. Y., Mckeever, E., Arras, P., & Weinberg, N. N. (2025). *Dynamical tide modified Roche limit in eccentric, asynchronous binaries*. Authors' version: https://arxiv.org/abs/2508.20183.

APPENDIX G

The physical environment table binding on the calculations

This appendix is the expanded version of the computational environment table presented at the front of the paper. Its purpose is practical: every calculation in the series is made for a particular moment and a particular place in the scenario, and the table records, quantity by quantity, what is known, what is adopted and what is still open at each stage. It prevents the most common error in reconstructing an ancient state — borrowing a value from the present Earth, or from a different stage, and inserting it where it does not belong. Each row is coded under a fixed identifier (E01, E02, …) so that later papers of the series can cite it.

How to read the table

Every row carries one of five statuses. **Adopted:** an input chosen as a working basis in the model. **Computed:** a result derived within particular assumptions. **Reproduced:** a calculation whose run code was executed and reproduced, without a commitment to validating all of its underlying assumptions. **Open:** a quantity not yet determined or not yet demonstrated. **Present constraint:** a measurable datum on today's system, used to limit the ancient reconstruction, and not a measurement of the past.

The columns of the table follow the scenario in time, and the stages describe one system, not three separate systems. **Stage A** — a reference state before the division of mass, when the lunar material is still counted in the Earth. **Stage B** — construction and transfer, while the body grows and rises. **Stage C** — a state close to release. All the data in column C refer to a defined coupled reference state, not to an evolutionary path already solved in full. Differentiation, a non-zero latitude or a mixing of material sources require a further consistent calculation.

The table is divided into thematic blocks: time, mass and geometry; the Earth's interior, rotation and angular momentum; gravity and where it is measured; heat, material, flow and the connection; and two later updates by the author on volatiles, cooling and the timeline.

The environment table — time, mass and geometry

This first block fixes the basic bookkeeping of the model: when the scenario takes place, how much mass there is and to which body each part is assigned, how large the bodies are, and where the source lies. Its main purpose is to prevent double counting — the same material cannot be included in both the Earth and the Moon — and to keep apart quantities belonging to different reference states.

ID	What is seen or measured	A — initial reference state	B — during construction	C — close to release	Status and meaning for the calculation
E01	The geological epoch	around 4.5 billion years ago	within the same estimated period of planetary formation	exact detachment age not yet determined	Chronological framework of the scenario; no exact dating of the proposed detachment event
E02	Elapsed time	$t = 0$ (the zero point is defined by the start of the transfer of mass)	time t advances; construction rate defined as open	t_{det} defined as open	A variable parameter; the 600 Myr working window is not a measured closure time, and a shorter alternative duration is also not yet determined
E03	Stage of the body and its evolution	lunar material still wholly assigned to the parent mass in the reference account	the bulge and the attached body grow with time	attached body up to the point of failure of the connection	The full lunar mass must not be applied to the initial bulge without stating explicitly that this is a limiting example
E04	Total system mass	6.0455×10^{24} kg	the mass is kept constant under the closed-system assumption	the same mass in the absence of loss	Adopted; the sum of the adopted model inputs; the value 6.045×10^{24} is rounding only
E05	Parent mass	6.0455×10^{24} kg	$M_E\,(\,t\,)$ decreases with net transfer	5.972×10^{24} kg in the base account	Excludes material already assigned to the developing body; material in the connecting element (the pear) requires explicit assignment

ID	What is seen or measured	A — initial reference state	B — during construction	C — close to release	Status and meaning for the calculation
E06	Mass of the developing body	no separate mass in the reference account	$m(t)$ grows as a function of the feed and incorporation rates	7.35×10^{22} kg as the base target	Adopted; the mass is derived from present lunar data under the loss assumption, and is not a direct measurement of the mass at detachment
E07	Material in transit and losses	wholly included in the parent budget	material moving along the pear — the connecting passage between the Earth's interior and the Moon — and material in ejection may form a separate reservoir	assignment and loss require closure	The same material must not be counted simultaneously in the parent and in the Moon; where needed, a separate third reservoir is defined

ID	What is seen or measured	A — initial reference state	B — during construction	C — close to release	Status and meaning for the calculation
E08	Source of the material	a deep interior reservoir and the outer shell that rises with the body	internal feed and external incorporation in parallel	share of each source open	The split 96%/4% (interior/shell) is a first estimate from the early shell picture of Paper 3, §B.7, and requires recomputation; the incorporation fraction is not measured. The Moon’s far-side crust is made of many layers of compacted sial that migrated to the tip of the pear, plus material that returned to it again and again; the thickness of the shell is not fixed in this block
E09	Volumetric radius of the parent	6,517 km	$R_V(t)$ varies as a function of mass, heat and structure	open in the general path	Adopted; the addition of 120 km above the computed cold radius (6,397 km) is an assumption, and is not an expansion relative to the present Earth’s radius at the same mass

ID	What is seen or measured	A — initial reference state	B — during construction	C — close to release	Status and meaning for the calculation
E10	Equatorial and polar radii	about 6,721 and about 6,128 km	the radii vary with spin and structure	equatorial radius of about 6,635 km in the reproduced $k = 0.4$ scenario	Based on a homogeneous Maclaurin figure in the reference approximation; these values must not be applied automatically to a body with a layered structure
E11	Parent density	mean value of about 5,214 kg/m³	evolving spatial and radial density profile	final profile defined as open	The mean density does not represent the local density at the feed depth
E12	Dimensions of the developing body	initial bulge radius defined as an open parameter	$r_b\,(\,t\,)$ varies with mass, heat and porosity	about 1,890 km, only if the whole target mass is at density 2,600	Reference values; a radius of about 2,270 km is obtained for the whole target mass at density 1,500 as a sphere; as a hemisphere on a base of 6,000 km, the same mass gives a density of 1,300 (Paper 3). These are two sizes of separate reference states, not a computed history

ID	What is seen or measured	A — initial reference state	B — during construction	C — close to release	Status and meaning for the calculation
E13	Distance between the centres of the bodies	undefined before a separate body is distinguished	$D(t)$ rises as a function of the lift mechanism	depends directly on the release scenario	The value 8,991 km is 6,721+2,270 in the geometric example of two touching bodies, and is not a necessary initial state
E14	Location of the source	the base of the pear stood on the Earth's surface near Kiritimati, 1.9°N 157.5°W, 9,000–11,000 km wide (Paper 3, Paper 9); the earlier reading around 25°N, 165°E in present geography is now defined, per the author, as the back of the pear (Paper 3)	a broad regional source area, not a mathematical point	latitude relative to the ancient spin axis open	Author's reconstruction; this location is not automatically equal to the dynamical release latitude, adopted as 22.5° in Paper 4
E15	Centres of mass	the centre of mass of the system is defined for all the mass	the centres of the two bodies and the centre of the system change with the migration of material	must be computed in the two-body system	It must not be assumed that the centre of today's Earth coincides with the centre of mass of the connected system

What this block establishes: one system mass, divided between the two bodies and, where needed, also a third reservoir of material in transit; reference sizes of radius and volume, belonging to stated densities and states; and a feed source that is a broad region, not a point.

The environment table — Earth's interior, rotation and angular momentum

The second block collects the quantities that set the spin and the budget: the distribution of mass inside the Earth (the inertia coefficient k, which is 0.4 for a uniform sphere and smaller when the mass is concentrated toward the centre), the moment of inertia, the period, the spin axis, the total angular momentum of the system and the corotation radius. These are the quantities of Parts A and B. The role of the block is to record which values belong to the initial reference state and which to the coupled state near release, and so to prevent the two being mixed.

ID	Quantity	A — initial reference state	B — during construction	C — close to release	Status and meaning
E16	Organisation of dense material	$k = 0.4$ (base assumption for the homogeneous reference)	iron sinks inward and the density organises in the scenario	k_f not known in advance	A body that has not completed the process of differentiation is not necessarily uniform
E17	Inertia coefficient k	0.4	$k\,(\,t\,)$ required for the calculation	0.4 in the base account; lower values in the sensitivity test	The values 0.3644 and 0.3307 are sensitivity inputs, not measurements of the Earth at that time
E18	Moment of inertia I	about 1.092×10^{38} kg m^2	varies as a function of mass, shape and density distribution	about 1.052×10^{38} in the coupled homogeneous scenario	$I = kMa^2$ here, where a is the equatorial radius; the definition of the coefficient k must be consistent
E19	Rotation period	5.5 hours adopted	T(t) derived from the angular-momentum balance and the structure of the body	5.77 hours at release (core centralising, $D = 2$)	Transfer of angular momentum to the orbit (slowing) and contraction/differentiation (speeding up) may act in opposite directions; a constant spin rate must not be assumed throughout the process

ID	Quantity	A — initial reference state	B — during construction	C — close to release	Status and meaning
E20	Local rotational speed	depends on the perpendicular distance from the spin axis	$v = \omega \varpi$ wherever corotation holds	the same definition relative to the axis at that moment	$\varpi = r \cos(\text{latitude})$; at non-zero latitude $v = \omega r$ must not be inserted automatically
E21	Spin axis	initial reference axis	may change its spatial direction through exchanges of angular momentum	open in the non-equatorial solution	The earlier value 4.48° was set apart (it needed a frame-consistency correction, Appendix D); the corrected values are 3.85° (test-particle approximation) and 3.84° (two-mass solution). These are algebraic outputs, not an approved final direction of a full dynamical system
E22	Total angular momentum	about 3.466×10^{34} kg m^2/s	continuous vector tracking; conserved in the absence of an external torque on the system	includes the spins of both bodies, their orbit and the additional material	A scalar-sum check does not constitute proof of full vector conservation
E23	Corotation radius	about 15,818 km in the 5.5-hour reference state	$R_{\text{sync}}(t)$ varies as a function of mass and spin rate	about 16,331 km at release (16,399 km in the two-body solution)	These are two separate approximations of one state. The value 17,895.9 is withdrawn. Reaching corotation is not in itself proof of detachment

ID	Quantity	A — initial reference state	B — during construction	C — close to release	Status and meaning
E24	Addition of angular momentum and holding in orbit	addition of about 15.6% in the reference account	depends on the source of the material and on the transfer rate	about 16.5% added and about 17.7% final orbital share, test particle, coupled with the pear alone ($D = 1$); 17.0% orbital share at release with $D = 2$ (§B.1)	The same term must not be used for the three percentages; the initial angular momentum of deep material and of shell material is not necessarily the same

What this block establishes: the spin rate and the corotation radius change during construction, in both directions; the two values of the corotation radius at release (16,331 km for a test particle and 16,399 km for two masses) are two approximations of one physical state; and the three percentages of E24 measure three different things.

The environment table — varying gravity and the location of measurement

The third block deals with gravity, which in this scenario is not one number but a quantity that varies in place and time. As mass passes from the Earth to the developing body, the pull at a given point changes, and it changes differently outside the Earth, at its surface, inside it and in the connection between the bodies. Each row names one place or component and states what a calculation made there must include. The purpose of the block is to prevent a gravity value valid at one place or stage from being used at another, and to ensure that gravity is derived from the actual mass distribution at that moment.

ID	Component or place	What happens during construction	What must be included in the calculation	Status
E25	Parent gravity at a fixed external point	the central mass component decreases as the parent mass falls	$M_E\,(\,t\,)$, the distance at that moment, and shape corrections if required	It must not be inferred automatically that surface gravity also falls

ID	Component or place	What happens during construction	What must be included in the calculation	Status
E26	Gravity on the Earth's surface	the mass decreases, but the local radius and the shape may change at the same time	location, local radius, density distribution and spin	Normal and tangential components must be separated and reported individually, not a single gravity number for the whole Earth
E27	Core organisation and the external field	a spherically symmetric internal distribution does not change the external field at the same distance, if the mass is fixed	effects of contraction, non-sphericity and spin must be computed separately	There is no 'additional gravity from the core' beyond the gravity of the same mass already counted in the budget
E28	The internal field	varies with the mass distribution and with the organisation of iron	density profile and internal potential; pressure along the path	GM_{total}/r^2 must not be used inside the Earth as if all the mass lay below the measurement point
E29	Gravity of the developing body	its contribution at the same distance grows as its mass grows	$m\,(\,t\,)$, radius and shape of the body; distance from the point	Growth of mass alone is not enough: the radius changes too
E30	The connection region	the attractions of the two bodies act in combination; pressure stresses and material strength act in addition	vector sum of the fields, geometry of the connection and motion of the frame	Cancellation of a force component at one point is not the same as the disappearance of stresses along a broad connection
E31	The external shell	moves on a surface that changes and curves	projection of the total field on the local tangent and normal	Gravity that holds material to the surface is not automatically gravity that drives closure
E32	Density anomaly Lobe A	in the scenario may change during the feeding and organisation of the material	density difference relative to the background, excess/deficit mass, depth and shape	Missing input; it must not be added to a model that already includes the same mass, so that it would be counted twice

ID	Component or place	What happens during construction	What must be included in the calculation	Status
E33	Centrifugal acceleration	varies with the spin rate and the distance from the spin axis	$\omega(t)^2 \varpi$, within a defined rotating frame	Not an additional gravity field; its direction is perpendicular to the spin axis, not necessarily radial from the Earth's centre
E34	Effective gravity	combination of gravity and the defined frame accelerations	in a varying field the motion of the origin must also be followed; in relative motion, Coriolis and Euler as needed	8.25 m/s² is an earlier reference estimate at a particular place, not an approved universal value for all stages of the scenario
E35	Tides and survival	the differences in attraction across the body vary with distance and with the masses	field gradient, body dimensions, density and material strength	The rigid/fluid reference limits of about 10,314/19,971 km for density 2,600 are reference bounds only, not a survival solution for a hot body
E36	The Sun and the orbital environment	may transfer torque and change the orbit	included in the calculation or explicitly neglected, according to the question and the duration	It must not be assumed to be negligible at every stage merely because it was neglected in a short estimate

What this block establishes: gravity must be derived from the actual mass distribution, at the stated place and moment; the centrifugal acceleration and the other frame accelerations (Coriolis and Euler) are added only within a stated rotating frame; and the Roche limits of E35 serve as reference distances only, while the question of survival is treated as the separate requirement of §D.3.

The environment table — heat, material, flow and the connection

The fourth block moves the focus from mechanics to the state of the material: temperature, heat, pressure, density, volatiles, viscosity, the outer shell and the connection, and what happens after release. These are the quantities that Paper 3 (*LEF Mechanism in Practice*) needs in order to test whether the material can rise, whether the shell can carry the load and whether the record on the

surface can be preserved. The block records the working values in use and, column by column, what is still to be fixed.

ID	Domain	The description required	Working value or state	What remains to be clarified
E37	Temperature	separate determination of the thermal conditions in the Earth's interior, along the feed path, in the shell, in the body and in the connection	no approved thermal profile	Temperature, gradient and time; 'hot' is not a number that can be inserted in an equation
E38	Heat sources and losses	internal separation, flow, deformation, latent heat and cooling	depends on the chosen scenario	Power and energy balance; a fully differentiated core must not be assumed as a heat source if the model requires it to be still organising
E39	Pressure	the pressure at the source, the pressure along the feed and the pressure in the connection	A source pressure of 150.26 GPa and a lift cost of 139.78 GPa, leaving a residual pressure of 10.49 GPa, are the reference numbers of Paper 3	Not measurements; they must be recomputed after any change of structure, spin or geometry
E40	State of the material	solid, partially molten, liquid, or material containing gas cavities	differs between regions of the system and changes in time	Melt fraction, composition and state law; not every 'hot material' is the same material
E41	Density of the rising material	depends on pressure, temperature, composition and gas	1,500 kg/m³ represents an adopted light/expanded body state	Does not prove a similar density at depth; the equation of state (EOS) at depth is open
E42	Volatiles and porosity	dissolved at depth or released through decompression	the quantities and the spatial distribution not yet determined	Time of cavity formation; the bulk density after freezing is not the density of the deep feed

ID	Domain	The description required	Working value or state	What remains to be clarified
E43	Rheology of the body and of the Earth's interior	viscous, elastic and plastic response by region and loading time	no approved rheological profile	The range 10^{18}–10^{20} Pa·s was proposed as a working estimate for survival assessments; it is not a proven result
E44	The sialic shell	the whole thickness carries stress; the outer part is colder and stronger, and the base hotter and more mobile	many layers of compacted sial that migrated to the tip of the pear, plus material that returned again and again; the shell thickness is not fixed; $\rho = 2,800$ kg/m³ kept as a reference input	Strength, viscosity, bending and tearing
E45	The interface layer	shear-stress transmission and rate of relative motion	thickness and viscosity open	A material law is required; the same resistance must not be counted both as friction and as viscosity without a combined model
E47	Geometry of the connection	a broad, hot region is possible, varying in area and thickness	$R_c = 500$–$1,500$ km proposed as a test input; $d_0 \approx 105$ km, set by the release-distance search of Paper 3 (an earlier version proposed $d_0 = 20$–50 km as test inputs only)	The proposed values must not be identified with a measurement of the ancient neck
E48	Failure of the connection	rounding, pressure, spin and the connection forces examined together	rounding fully completed before release (author, 6.10.2026)	At release the outer shell is rigid and round. No assumption of a narrow pinch is needed
E49	Lift velocity	sets the power, the shear stresses and the release velocity	$v_r\,(\,t\,)$ defined as open	Not derived from the corotation identity or from basalt ages

ID	Domain	The description required	Working value or state	What remains to be clarified
E50	Closure conditions	balance between supply, attraction, resistance and conservation of mass	the outer shell may close slowly; its closure plays no part in the order of detachment	Mobility, time and preservation must be shown within the same parameter domain
E51	Preservation of the surface	a record can rise with the shell if it survives the deformation and heating	conditional prediction	A strain of 3% and a viscosity of 10^{23} Pa·s are not a preservation calculation reproduced in the last round
E52	The external cooling environment	heat loss by radiation, a possible atmosphere, vapour and water according to the epoch	not determined	The present Earth's surface conditions or early liquid water must not be assumed without basis; required only for the relevant thermal solution
E53	After release	no further continuous deep feed from the Earth; the body passes to a free orbit	initial conditions from state C required	The angular momentum of the two spins and of the orbit, re-contact and survival must be checked; late volcanism on the Moon is not a continuation of the feed from the Earth's interior through the pear

What this block establishes: the mechanical working values that Paper 3 uses as input (the pressures of E39, the shell of E44 and the connection of E47); the fixed point that at release the Moon is a rigid, round body (E48); and an explicit list of thermal and material quantities still to be determined. The flattening and rounding of the body took place earlier, while it was still attached (Paper 7, *The Poles*).

The card to accompany every calculation

The table above becomes a binding working rule: a short card that every calculation in the series states before its result, so that the reader sees at once which state it describes, and so that data are not borrowed from the present or from another stage.

Before an equation or a table of results, state: (1) stage and time — the stage of the scenario (A, B or C) and the time t; (2) point of measurement — external, at the surface, internal or in the connection region; (3) frame and axis — inertial or rotating, and the position of the spin axis; (4) scenario version — for example a homogeneous or layered profile; (5) masses and radii valid at that stage; (6) heat and material inputs — thermal parameters, melt fractions and viscosity; (7) effects neglected — for example the solar perturbation; (8) result and sensitivity. If a datum is missing or not yet determined, the calculation is marked explicitly as conditional or deferred — a value must not be borrowed from the present or from another stage without justification.

The mass bookkeeping is written as one equation. Mass-management equation: $M_{\mathrm{total}} = M_E + m_{\mathrm{body}} + M_{\mathrm{transit}} + M_{\mathrm{escaped}}$, according to the boundaries of the system and the reservoirs: the Earth, the developing body, the material on its way between them (in the pear), and any material lost from the system. If a closed system without loss is chosen, or the material of the connection is assigned wholly to one of the two bodies, this must be stated explicitly in the card and the terms of the equation reduced accordingly. The mass entering GM must match the reservoir definition in the same calculation.

The combined gravitational field must be derived from the actual mass distribution, at the stated moment and place. A two-point-mass approximation is permitted only in the domain where it was defined and at the precision defined for it; it is not a full description of the Earth’s interior, of the connection region or of a local anomaly. In a decomposition into a background component and a local anomaly component (such as Lobe A), only the difference mass is added, so that no mass is counted twice.

The author’s update — the ancient volatile inventory and the returning material

This update adds to the table the volatiles — water, gases and other substances that evaporate easily — which affect the density, the flow, the cooling and the returning material. It records the author’s picture and the distinctions that any later calculation must keep, without computing quantities at this stage.

Explicit instruction from the author: in the scenario examined, the ancient volatile inventory is far larger than the inventory remaining today. Volatiles are present in the material from which the system is built, including the deep material and the magma; part of the volatiles that passed with the developing body were lost to space, and according to the hypothesis further loss occurred during detachment. There is no request at present to compute the quantities or the loss rates.

Additional ID	Quantity/distinction	Binding treatment
E54	Ancient volatile inventory	An initial inventory term higher than the inventory remaining today, per the author’s assumption; the initial quantity must not be derived directly from present samples or measures
E55	Location of volatiles	The description includes the deep material, the magma, the developing body and the ejected material; the composition and quantity profile in each reservoir remain open
E56	State of the volatiles	The author does not initially describe a reservoir of liquid water. A component dissolved in the silicate solvent, a separate phase, bubbles and vapour must be distinguished as a function of pressure and temperature; gas bubbles must not be assumed at every depth
E57	Loss during transfer and detachment	A proposed loss path, not quantified. Exit from the solvent, emission from the body’s surface, residence in orbit and escape from the Earth–Moon system must be distinguished; evaporation alone is not proof of escape
E58	The mass account	If the loss is not negligible, a separate reservoir of lost volatiles must be added to the mass-management equation. The value 6.0455×10^{24} kg remains the reference mass without loss, and is not a closed estimate of all the initial mass in a scenario with loss
E59	The returning material	The author proposes a link between the volatiles and density relief on one side and the returning material and crater formation on the other. At this stage this is a claim to be tested; cratering capacity must not be derived from density or viscosity data alone, and the calculations of Paper 6 are not to be opened now
E60	Viscosity versus density	The author wrote ‘high viscosity’ and ‘lightened the material’. His words are to be preserved, and it is to be clarified whether a viscous magma holding bubbles is meant, or low density/easy flow. Resistance to flow (viscosity) and lightness (density) are two different properties: high viscosity is not equivalent to low density. Until this is clarified, no numerical causal link between them is fixed

The rows separate things that are easily run together: how much volatile material there was (E54), where it was (E55), in what form (E56), how it was lost (E57), how the loss enters the mass account (E58), and how it may relate to the returning material that later lands on the Moon (E59). E60 separates two properties of a magma — resistance to flow and lightness — which a magma rich in gas bubbles may show at the same time, but which are not the same physical property.

Cumulative update — volatile cycle, cooling and the timeline

The second update describes how the young system cooled while it was being built, and fixes the sequence of events from active construction to a separate body.

11 September 2026. This update is added to all the previous content and extends the environment table; it does not replace tables E01–E60 or the distinctions in them. The document now contains 70 items. The earlier reference to 53 quantities describes the original base table only.

Author’s clarification: in the proposed picture of the environment, the water inventory represented today in the seas and oceans was, at the start of the scenario, in a vaporised state, and the system was rich in volatiles also within the deep material. A cyclic motion takes place: volatile-rich material rises, gives off heat and cools, and part of it sinks back, is heated and rises again — like a pot of soup on a stove, where the surface gives off heat while the interior keeps turning over. According to the hypothesis, the cycle assists the cooling of the outer surface and the preservation of the record, while the interior remains hot and mobile.

Status: this is the description of the scenario that the author asks to put to the test. It contains no verified quantitative determination of the ancient water inventory, of the state of the volatiles at every depth, of the duration of construction or of the cooling capacity. The previous distinction between assumption, result and datum is preserved in full. The 600 Myr window must not be replaced by another short time without an explicit decision, and must not be presented as a measured closure duration.

The following table enters the cycle into the picture of the environment row by row: the water at the start (E61), the cycle itself (E62–E64), its effect on the shell (E65), the time window (E66), and the decline of the outward ejections, their end at the moment of detachment and their relation to the internal feed (E67–E70).

Additional ID	Domain	What enters the environment picture	Definition and status for the calculations
E61	Water inventory at the start of the scenario	Per the author, the water represented today in the seas and oceans was then in a vaporised state	Scenario assumption; the division of the ancient inventory among the atmosphere, the magma and the deep reservoir remains open. The present quantity of water must not be used automatically as a uniform ancient concentration in all the material
E62	Cycle of motion and cooling	Rise of volatile-rich material, cooling, sinking back and reheating	A proposed convection mechanism. Which material returns, and in which phase, must be defined; it is not automatically the same as the ballistic returning material that forms craters
E63	Condensation and return of material	Part of the material may condense on reaching suitable conditions and sink back downward	A possibility within the cycle, not an assumption that all the material returns as liquid rain or that the whole atmosphere condenses at the same stage
E64	Heat exit	Heat is transferred through the atmosphere and finally radiated to space	The path required in the model. The net rate of heat loss is open; internal convection in itself is not a loss of energy from the system
E65	Cooling of the shell and preservation	The cycle may help the outer part cool and stiffen, while the base remains hot and mobile	A conditional prediction, dependent on the thermal balance, the loading time and the rheology. It must not be declared that preservation of the record is proven by the mere existence of the cycle

Additional ID	Domain	What enters the environment picture	Definition and status for the calculations
E66	The cooling window	About 600 million years in the author’s working scenario	‘600 years’ was clarified in the context of the conversation as 600 million years. This is a window for testing, not a crust/basalt age difference that proves a construction duration
E67	Decline of the ejections	The ejections continue as long as the Moon is attached to the Earth and subject to its internal pressure; they decline as the pear grows and the Moon moves away from the Earth’s centre, but they do not stop	A proposed causal link; there is not yet a quantitative law for the ejection rate
E68	Overlap of processes	Outward ejection of material and the rise of the dome take place in parallel during part of the construction period	The ejection rate and the rate of rise must be followed separately; it must not be assumed that the rise begins only after the ejections end
E69	End of the ejections	At the moment of detachment (stage D) the Moon is cut off from the Earth’s internal pressure, and the outward ejections stop	The sequence in the author’s scenario: ejections, steadily declining, for as long as the Moon is attached; their end at the moment of detachment
E70	Separation of ejection and feed	The end of outward ejection does not indicate that all transfer of material from inside to the body ended at the same moment	Three separate rates must be defined: internal feed, outward ejection and the rise of the shell. The time at which the internal feed stops is not determined in this update

Timeline for the description — without invented intermediate times

The timeline orders the stages by causal sequence, without assigning dates to the intermediate stages, since those dates have not been computed. Read each row across: it gives, for one stage, the state of the outward ejections, of the connection and of the heat. Read a column from top to

bottom to follow one process from its start to its end: the ejections decline as the pear grows, but continue as long as the body is attached, and stop at the moment of detachment.

Stage	Outward ejections	The connection	Thermal state per the scenario
A — active construction	present	body attached	A hot, volatile-rich system; a cycle of motion and cooling through the atmosphere
B — the pear grows	fading and declining as the pear grows and the Moon moves away from the Earth’s centre, but not ceasing	body still attached	Cooling processes continue; rate open
C — close to release	still present, at a lower rate	body still attached	Sufficient mobility required despite the cooling; no assumption that the whole system froze
D — detachment	stop: the Moon is cut off from the Earth’s internal pressure	detachment takes place at a time not yet computed	Rounding fully completed before release, in accordance with E48
E — separate body	no continuous deep feed from the Earth	body separate	Late lunar volcanism and continued cooling are separate processes, and do not contradict the end of the ejections at the moment of detachment

How the update connects to the previous tables

The list below ties the update back to the rows it touches, so that a reader following any one quantity can find every place where it appears.

- E01–E02 (the time window): the scenario window and the epoch are kept separately, without mixing them with stage times not yet determined.
- E37–E38 and E52 (the thermal profile): the volatile cycle and the path of heat loss by radiation were added to the thermal description; the temperature and power values are still open.
- E42 and E54–E58 (the volatile balance): the inventory, the phase, the release and the escape are kept as separate quantities; dissolved volatiles must not be converted into free vapour at every depth.

- E44–E46 and E50–E51 (the structure of the shell): the graded structure of the whole thickness of the shell, the mobility and the preservation of the record are kept. The proposal of a cooling cycle does not cancel the requirement of the mechanical test.
- E48 and E53 (the release): rounding is completed before release, and the distinction between an attached body and a body in orbit after release is kept.
- E60 (rheology): the question of viscosity versus density has not been deleted, and is not resolved automatically by the clarification of the vapour cycle.

For future examination, without opening a calculation now: heat flux through the vapour atmosphere; latent heat in the stages of evaporation and condensation; the effect of the atmosphere on the emission of radiation; the balance of water and volatiles among the reservoirs; and the consistency among cooling, duration of activity, mobility of the shell and preservation of the record. There is no determination here that a vapour atmosphere necessarily accelerates cooling under all conditions.

Use of AI tools: The author used AI assistants (Claude, Anthropic; ChatGPT, OpenAI; Google's AI search assistant) to write and run the computation scripts, to analyse the catalogues and maps, to prepare figures and tables, to discuss the scenarios, and for language editing and translation. The model and its conclusions are the author's, who checked the content and takes full responsibility for it.

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